CHAPTER 05

Alcohols

Content and rendering review packet

Original app content and actual Chromium screenshots. No textbook passages or textbook figures are embedded. Source labels and curriculum headings are not evidence of factual correctness.

Source snapshot SHA-256: 124e2028ab3e14ac54d07077703f0e06e986a19840a9cf9d282f899a55cc637e

Review all claims, molecule graphs, lessons, exercises, hints and displayed states. Keys are author claims, not ground truth. Screenshots show static states with reduced motion: they do not prove animation timing or persistence behavior.

For every finding: item ID/page; severity (critical/major/minor); exact claim; derivation or reference evidence; proposed correction; confidence. List unreviewed/uncertain items. Model agreement is not scientific verification.

Coverage checklist

ch05:exercise:classify-alcohol | classify-alcohol:stage-1, classify-alcohol:stage-2, classify-alcohol:stage-3

ch05:exercise:oxidize-secondary | oxidize-secondary:stage-1, oxidize-secondary:stage-2, oxidize-secondary:stage-3

ch05:exercise:protection | protection:stage-1, protection:stage-2, protection:stage-3

ch05:exercise:quinone-redox | quinone-redox:stage-1, quinone-redox:stage-2, quinone-redox:stage-3

ch05:lesson:properties | properties:step-1, properties:step-2

ch05:lesson:properties:rule | properties:rule

ch05:lesson:properties:cue | properties:cue

ch05:lesson:properties:trap | properties:trap

ch05:lesson:properties:app | properties:app

ch05:lesson:oxidation | oxidation:step-1, oxidation:step-2, oxidation:step-3

ch05:lesson:oxidation:rule | oxidation:rule

ch05:lesson:oxidation:cue | oxidation:cue

ch05:lesson:oxidation:trap | oxidation:trap

ch05:lesson:oxidation:app | oxidation:app

ch05:lesson:activation | activation:step-1, activation:step-2, activation:step-3

ch05:lesson:activation:rule | activation:rule

ch05:lesson:activation:cue | activation:cue

ch05:lesson:activation:trap | activation:trap

ch05:lesson:activation:app | activation:app

ch05:lesson:phenols | phenols:step-1, phenols:step-2, phenols:step-3

ch05:lesson:phenols:rule | phenols:rule

ch05:lesson:phenols:cue | phenols:cue

ch05:lesson:phenols:trap | phenols:trap

ch05:lesson:phenols:app | phenols:app

ch05:molecule:isopropanol | isopropanol

ch05:molecule:tbutanol | tbutanol

ch05:molecule:acetone | acetone

ch05:molecule:acetaldehyde | acetaldehyde

ch05:molecule:mesylate | mesylate

ch05:molecule:acetal | acetal

ch05:molecule:phenol | phenol

ch05:molecule:hydroquinone | hydroquinone

ch05:molecule:quinone | quinone

ch05:curriculum | scope-1, scope-2, scope-3

Curriculum scope

scope-1: 5.1 Description and Properties

scope-2: 5.2 Reactions of Alcohols

scope-3: 5.3 Reactions of Phenols

Objective: Classify alcohols and physical properties

Objective: Predict oxidation and protection outcomes

Objective: Distinguish phenols, quinones, and hydroquinones

Supplied Kaplan curriculum: Chapter 5; section scope checked against local body and summary | organic_chemistry_kaplan/OEBPS/chapter05b.xhtml

IUPAC Gold Book, acetals (search-index definition inspected; direct page fetch unavailable) | https://goldbook.iupac.org/terms/view/A00062

OpenStax 17.2, alcohol properties | https://openstax.org/books/organic-chemistry/pages/17-2-properties-of-alcohols-and-phenols

OpenStax 17.6, alcohol reactions | https://openstax.org/books/organic-chemistry/pages/17-6-reactions-of-alcohols

OpenStax 17.7, alcohol oxidation | https://openstax.org/books/organic-chemistry/pages/17-7-oxidation-of-alcohols

NIH ODS, vitamin K | https://ods.od.nih.gov/factsheets/VitaminK-HealthProfessional/

PubChem BioCyc human ubiquinol-10 pathway, indexed description | https://pubchem.ncbi.nlm.nih.gov/pathway/BioCyc%3AHUMAN_PWY-5872

No source passage is included. Request relevant authoritative excerpts when a claim cannot be checked by derivation.

properties | Alcohol classes and physical properties

Classify the carbon bearing OH

properties:rule | Core Rules

A primary, secondary, or tertiary alcohol has one, two, or three carbon substituents on the OH-bearing carbon. Hydrogen bonding raises boiling points relative to comparable nonpolar molecules. Small alcohols mix well with water; increasing hydrocarbon size lowers water solubility. For comparable molecular sizes, additional OH groups strengthen hydrogen bonding; branching often lowers boiling point, but competing effects require actual data. In water, simple alkyl alcohol acidity generally decreases methanol > primary > secondary > tertiary: alkyl donation and poorer solvation destabilize alkoxide. Electron-withdrawing groups can lower pKa; solvent matters.

Visual card: Alcohol Physical Properties: Intermolecular Hydrogen Bonding, Boiling Point & Acidity

Chemistry diagram: Compare the carbon neighbors of the OH-bearing carbon. These are different illustrative alcohols.

O–H···O H-Bonding → ↑↑ BP

Chain ≤ 3–4 C = Water Miscible

Branching → ↓ BP (Less Surface)

Acidity: 1° > 2° > 3° ROH (Gas vs Solv)

Alcohol (ROH) | H-bond DONOR + ACCEPTOR | Highest BP vs ether/alkane of same MW

Ether / Ketone | H-bond ACCEPTOR only (Dipole) | Moderate BP; soluble in water only if small

Alkane (R–H) | London dispersion forces only | Lowest BP; insoluble in water

Boiling Point Hierarchy (Same MW) — — Carboxylic Acid (dimer H-bonds) > Alcohol (O–H H-bonds) > Ketone/Aldehyde (strong dipole) > Ether > Alkane (dispersion only).

Chain Length & Branching Effects — — Longer chains ↑ BP (more van der Waals surface) and ↓ water solubility; branching makes molecules spherical → ↓ surface area → ↓ BP.

Alcohol Acidity in Solution (pKa 15.5–18) — — In water, 1° ROH (pKa ~16) is more acidic than 3° ROH (pKa ~18) because bulky alkyl groups block solvent H-bonding around the RO^- alkoxide!

properties:cue | Memory Cues

Count C neighbors of C–OH; do not count the number of OH groups. Name a principal alcohol with -ol (or -diol for two OH groups); use hydroxy- when OH is a prefix. On benzene, ortho, meta, para mean 1,2-, 1,3-, 1,4-.

Visual card: Boiling Point & Solubility Predictor

Which physical property comparison is on the exam?

Butan-1-ol vs. Diethyl Ether (Both $C_4H_{10}O$) — Butan-1-ol Boils ~83 °C Higher (118 °C vs 35 °C) — Alcohols have an O–H bond that both donates and accepts intermolecular H-bonds; ethers lack O–H bonds and cannot H-bond to themselves.

Butan-1-ol vs. tert-Butanol (Chain vs Branched) — Linear Butan-1-ol Has Higher BP; t-BuOH Has Higher MP — Linear chains pack more surface area for dispersion in liquids (↑ BP), while spherical symmetry packs tightly into crystals (↑ MP).

Choosing a Base to Deprotonate ROH (pKa ~16) — Use NaH, Na/K Metal, or LDA (NOT NaOH Residual 100%) — NaOH (H_2O pKa 15.7) gives only an equilibrium mixture; NaH (H_2 pKa 35, evolves gas) drives alkoxide formation to 100%.

properties:trap | Common Mistakes / MCAT Traps

A secondary alcohol is not an alcohol with two hydroxyl groups. Phenol has OH directly attached to an aromatic ring and differs in acidity from an aliphatic alcohol.

Visual card: Thinking Ethers Can Hydrogen-Bond to Themselves

Common Trap — — Ranking diethyl ether near butan-1-ol in boiling point because "both have an oxygen atom that hydrogen-bonds".

Correct Reasoning — — Pure ethers have NO O–H, N–H, or F–H bonds: they can accept H-bonds from water, but pure liquid ether has only weak dipole-dipole forces.

properties:app | MCAT Applications

An OH group can increase water affinity and supply a hydrogen-bond donor and acceptor in a binding site, but whole-molecule properties still matter.

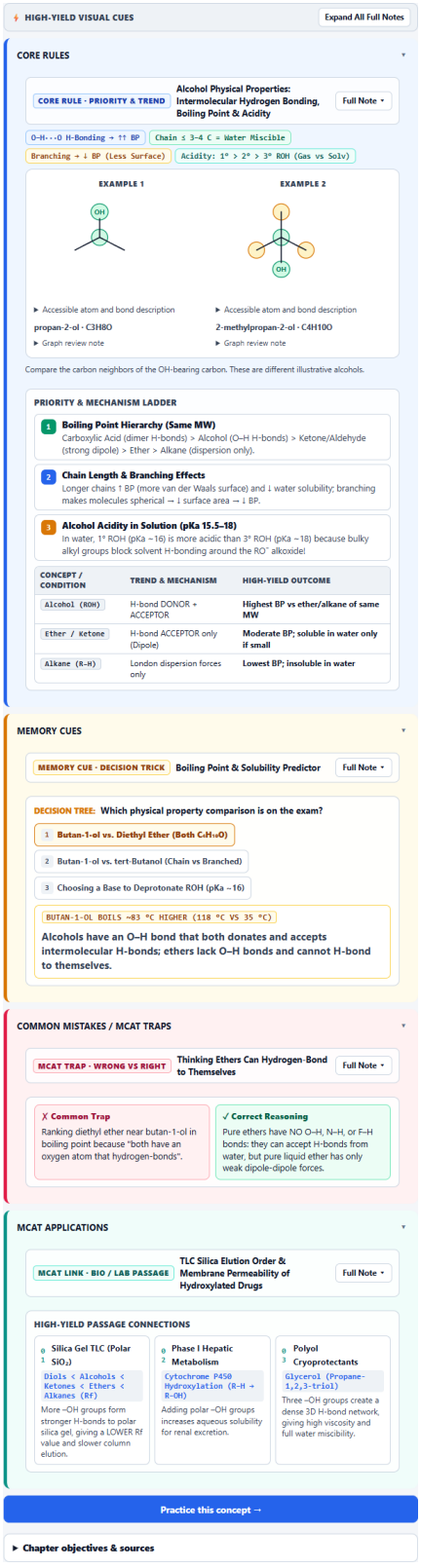
Visual card: TLC Silica Elution Order & Membrane Permeability of Hydroxylated Drugs

Silica Gel TLC (Polar SiO_2) — Diols < Alcohols < Ketones < Ethers < Alkanes (Rf) — More –OH groups form stronger H-bonds to polar silica gel, giving a LOWER Rf value and slower column elution.

Phase I Hepatic Metabolism — Cytochrome P450 Hydroxylation (R–H → R–OH) — Adding polar –OH groups increases aqueous solubility for renal excretion.

Polyol Cryoprotectants — Glycerol (Propane-1,2,3-triol) — Three –OH groups create a dense 3D H-bond network, giving high viscosity and full water miscibility.

Screenshot: properties-visuals



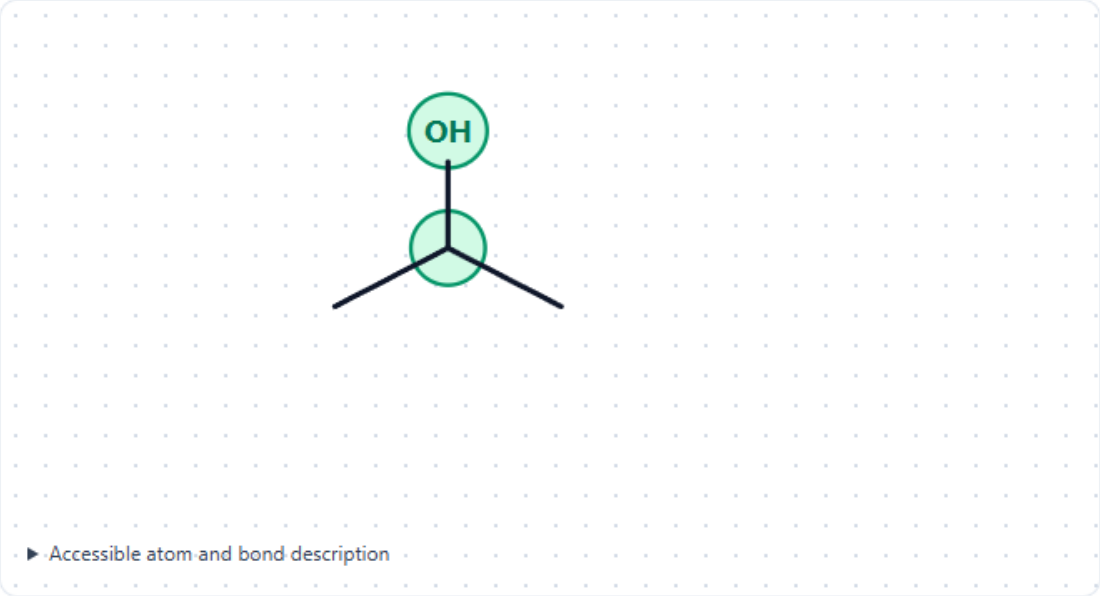
properties:step-1 | 1. Secondary alcohol

The OH-bearing carbon has two carbon neighbors and one H. It is a secondary alcohol.

Displayed molecule ID: isopropanol

{"highlights": {"prin": [1, 3]}, "revealName": true}

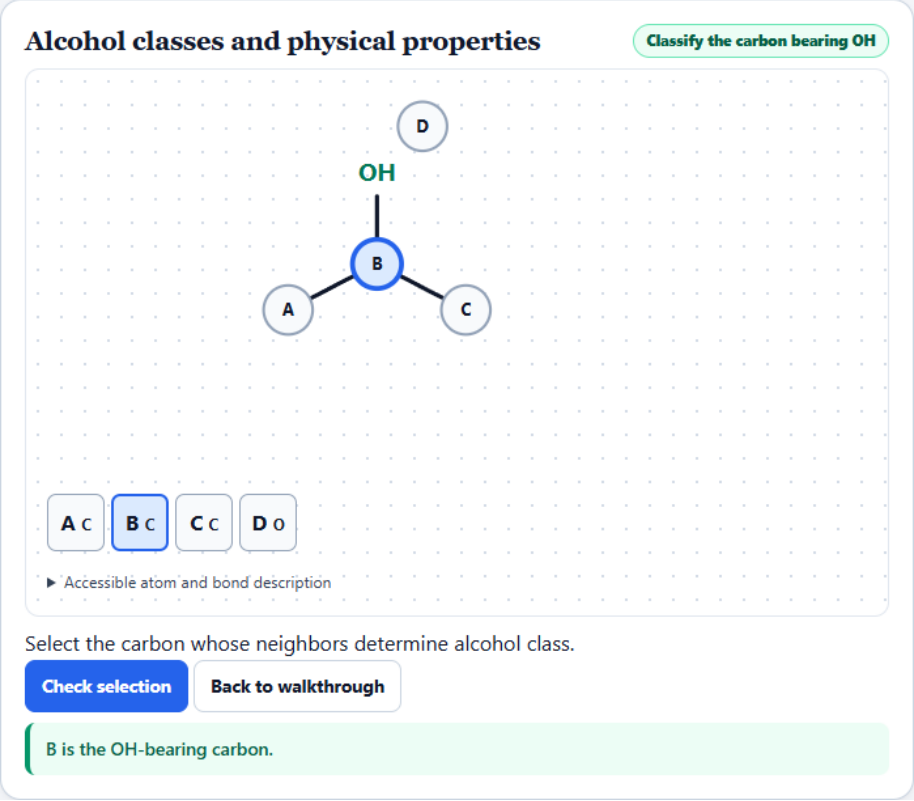
Screenshot: properties-step-1



Embedded challenge: Select the carbon whose neighbors determine alcohol class.

{"prompt": "Select the carbon whose neighbors determine alcohol class.", "answer": [1], "explanation": "B is the OH-bearing carbon."}

Screenshot: properties-step-1-challenge



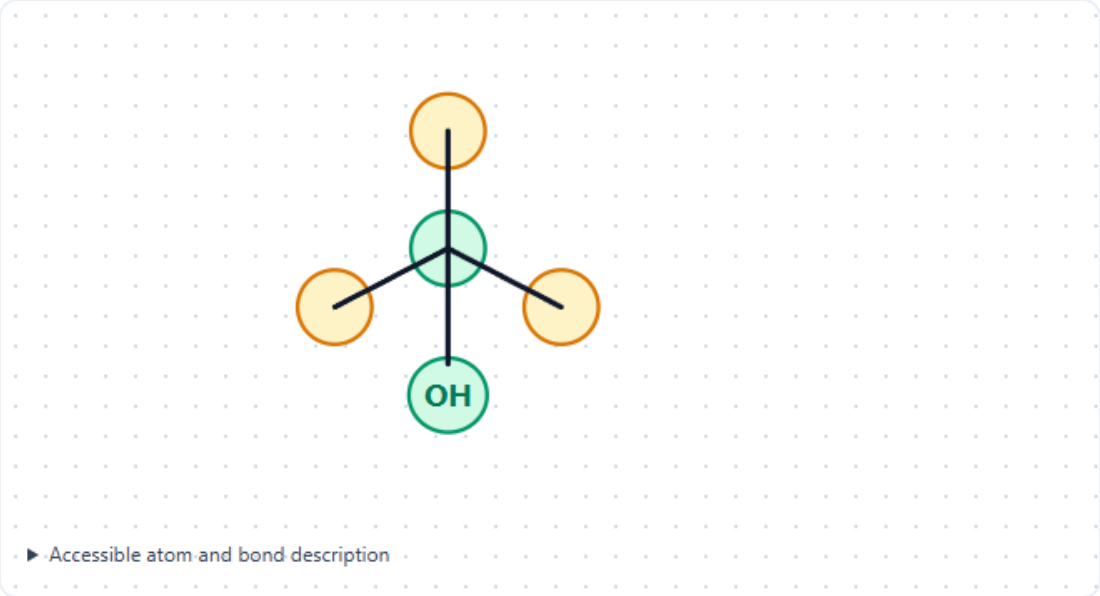
properties:step-2 | 2. Tertiary alcohol

Three carbon groups surround the OH-bearing carbon, which therefore has no C–H bond. This changes its usual oxidation behavior.

Displayed molecule ID: tbutanol

{"highlights": {"prin": [1, 4], "sub": [0, 2, 3]}, "revealName": true}

Screenshot: properties-step-2



oxidation | Predict alcohol oxidation products

Alcohol class + oxidant + conditions

oxidation:rule | Core Rules

Primary alcohols give aldehydes with suitable mild anhydrous oxidation (PCC is one example), or acids with suitable aqueous strong oxidation. Secondary alcohols give ketones. Tertiary alcohols do not undergo the ordinary C–H-removing oxidation without C–C cleavage. PCC (pyridinium chlorochromate) under anhydrous conditions can stop primary-alcohol oxidation at an aldehyde. Jones reagent, CrO3 in aqueous sulfuric acid, or acidified dichromate commonly gives a carboxylic acid. Water allows aldehyde hydrate formation and further oxidation; reagent and conditions govern selectivity.

Visual card: Chemoselective Alcohol Oxidation: PCC / Swern / DMP vs. Jones / $KMnO_4$

Chemistry diagram: Secondary alcohol to ketone with a suitable oxidant; schematic, not a balanced equation.

1° + PCC → Aldehyde

1° + CrO_3/H_2O → Carboxylic Acid

2° + Any Cr(VI) → Ketone

3° → No Reaction

PCC / DMP (Anhydrous) | 1° ROH → Aldehyde (RCHO) | 2° ROH → Ketone; 3° ROH → N.R.

Jones ($CrO_3/H_2SO_4/H_2O$) | 1° ROH → Carboxylic Acid (RCOOH) | 2° ROH → Ketone; Orange Cr^{6+} → Green Cr^{3+}

$NaIO_4$ (Periodate) | Vicinal 1,2-Diol → 2 Carbonyls | Cleaves C–C bond between adjacent –OH groups

Count α-Hydrogens on the Carbinol Carbon (C–OH) — — Each oxidation step to a C=O π bond removes ONE H from oxygen and ONE H from the attached carbon (1° has 2 H; 2° has 1 H; 3° has 0 H).

Stop at Aldehyde (Anhydrous Oxidants) — — PCC (pyridinium chlorochromate in CH_2Cl_2), Dess-Martin periodinane (DMP), or Swern ($DMSO/(COCl)_2/Et_3N$) stop cleanly at the aldehyde.

Over-Oxidation to Carboxylic Acid (Aqueous Strong Oxidants) — — Jones reagent ($CrO_3/H_2SO_4/H_2O$), $Na_2Cr_2O_7/H_2SO_4$, or hot $KMnO_4$ oxidize 1° alcohols all the way to carboxylic acids via the aldehyde hydrate.

oxidation:cue | Memory Cues

1° → aldehyde → acid; 2° → ketone; 3° lacks the necessary C–H.

Visual card: Instant Alcohol Oxidation Outcome Selector

What alcohol class and reagent are paired in the reaction?

Primary (1°) Alcohol + PCC / CH_2Cl_2 — Aldehyde (RCHO) — Stops Cleanly! — Anhydrous conditions prevent hydrate formation, preserving the aldehyde without over-oxidation.

Primary (1°) Alcohol + CrO_3 / H_2SO_4 / H_2O — Carboxylic Acid (RCOOH) + Green Cr^{3+} — Water hydrates the intermediate aldehyde to $RCH(OH)_2$, which is immediately oxidized a second time to RCOOH.

Secondary (2°) vs. Tertiary (3°) Alcohol — 2° → Ketone (With PCC OR Jones); 3° → No Reaction — Secondary alcohols have exactly 1 carbinol H (always yield a ketone); tertiary alcohols lack a carbinol H and fail the chromate color test.

oxidation:trap | Common Mistakes / MCAT Traps

PCC is not the only aldehyde-selective oxidant. Do not label every reagent containing oxygen “strong enough” or assume every tertiary alcohol is inert under all conditions.

Visual card: Using Aqueous Dichromate to Make an Aldehyde or Oxidizing a Tertiary Alcohol

Common Trap — — Reacting butan-1-ol with $Na_2Cr_2O_7/H_2SO_4/H_2O$ expecting butanal, or oxidizing 2-methylpropan-2-ol to a ketone.

Correct Reasoning — — Ordinary alcohol oxidation gives a ketone from a secondary alcohol. Primary-alcohol products depend on reagent and water; tertiary alcohols lack a carbinol H for this pathway, although harsher oxidative cleavage is possible.

oxidation:app | MCAT Applications

Alcohol dehydrogenases catalyze related biological redox steps using cofactors; substrate and cofactor specificity control the result.

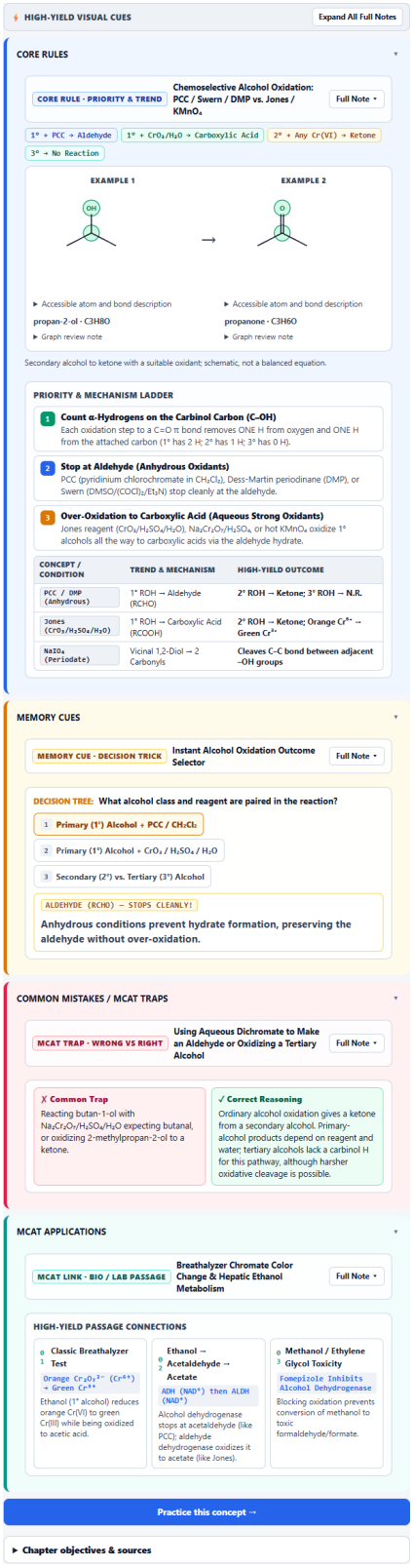
Visual card: Breathalyzer Chromate Color Change & Hepatic Ethanol Metabolism

Classic Breathalyzer Test — Orange $Cr_2O_7^{2-}$ (Cr^{6+}) → Green Cr^{3+} — Ethanol (1° alcohol) reduces orange Cr(VI) to green Cr(III) while being oxidized to acetic acid.

Ethanol → Acetaldehyde → Acetate — ADH (NAD^+) then ALDH (NAD^+) — Alcohol dehydrogenase stops at acetaldehyde (like PCC); aldehyde dehydrogenase oxidizes it to acetate (like Jones).

Methanol / Ethylene Glycol Toxicity — Fomepizole Inhibits Alcohol Dehydrogenase — Blocking oxidation prevents conversion of methanol to toxic formaldehyde/formate.

Screenshot: oxidation-visuals



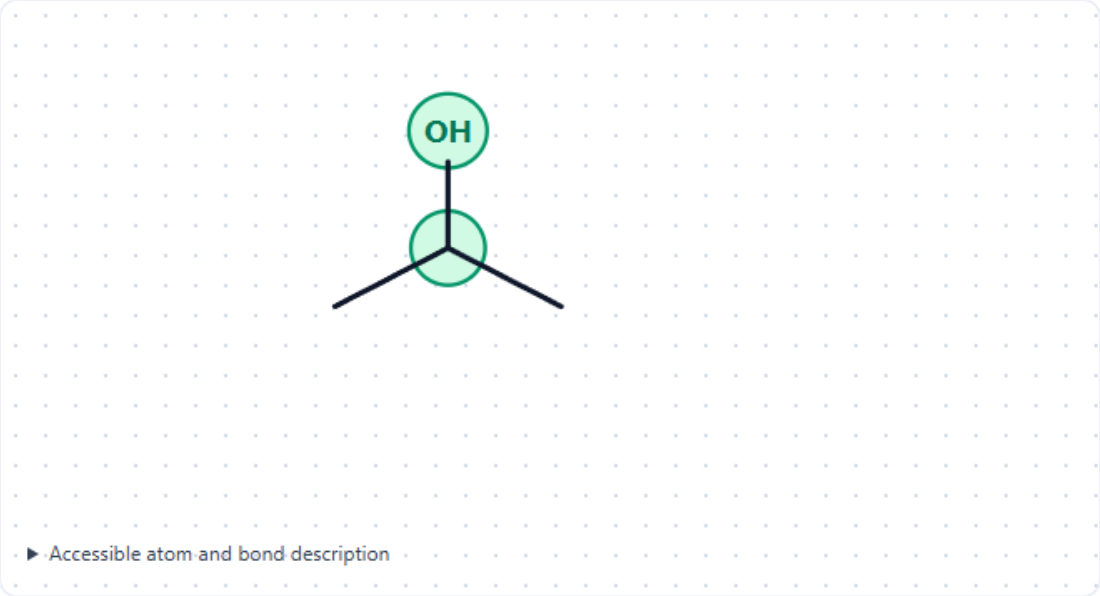
oxidation:step-1 | 1. Locate the oxidized carbon

Oxidizing propan-2-ol removes an O–H hydrogen and the H on its central carbon, producing C=O.

Displayed molecule ID: isopropanol

{"highlights": {"prin": [1, 3]}, "revealName": true}

Screenshot: oxidation-step-1



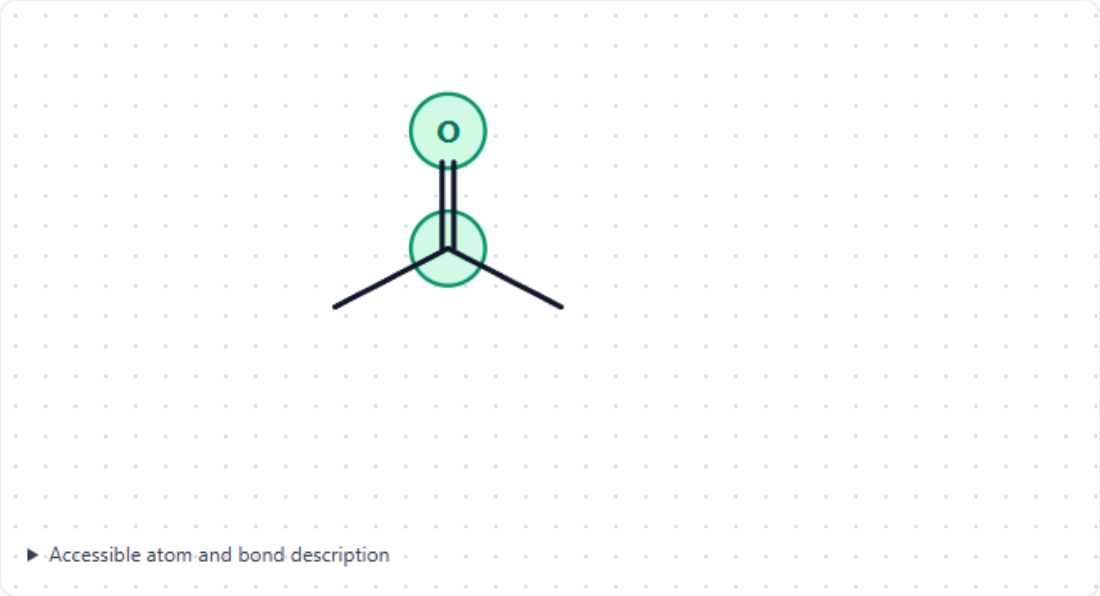
oxidation:step-2 | 2. Compare the product

The same three-carbon framework is now a ketone. Carbon–carbon connectivity is unchanged in this ordinary alcohol oxidation.

Displayed molecule ID: acetone

{"highlights": {"prin": [1, 3]}, "revealName": true}

Screenshot: oxidation-step-2



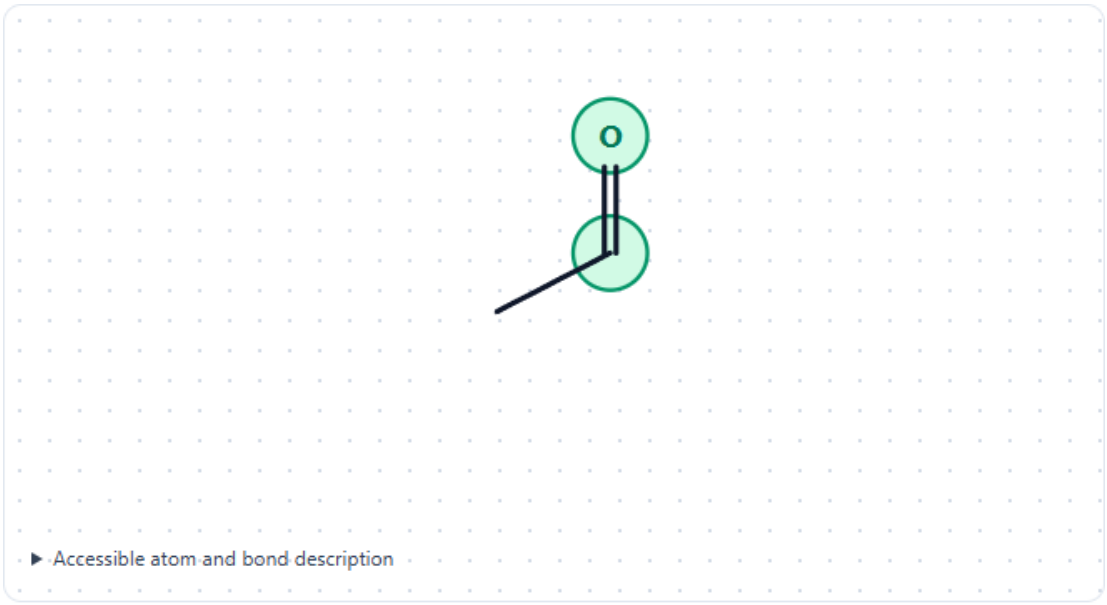
oxidation:step-3 | 3. Primary-alcohol conditions

Ethanal can be obtained from ethanol with an appropriate anhydrous oxidant. In aqueous strongly oxidizing conditions it can be oxidized further to ethanoic acid.

Displayed molecule ID: acetaldehyde

{"highlights": {"prin": [1, 2]}, "revealName": true}

Screenshot: oxidation-step-3



activation | Activate and protect functional groups

Make leaving groups; mask carbonyls

activation:rule | Core Rules

Converting ROH to a mesylate (ROMs) or tosylate (ROTs) gives a much better leaving group. The alcohol oxygen remains bonded to carbon during sulfonate formation. A later SN2 displacement at that carbon inverts its geometry. Acetals protect carbonyls under many basic conditions. MsCl or TsCl with a suitable base converts ROH to ROMs or ROTs. Acetal/ketal formation masks a carbonyl; these groups tolerate many bases and hydride reagents, including LiAlH4 under suitable conditions. Sulfonates are activated leaving groups, not universally inert alcohol protecting groups.

Visual card: Alcohol Activation (Tosylates, Mesylates, $SOCl_2$, PBr_3) & Silyl/Acetal Protecting Groups

Chemistry diagram: Separate leaving-group activation and carbonyl-protection examples; these are not a reaction pair.

TsCl/Pyr → Retention (1 Step)

TsO^- + Nu^- → Inversion

$SOCl_2$ / PBr_3 → Inversion

TBDMS / Acetal = Protecting Group

TsCl, pyridine | ROH → ROTs (RETENTION of R/S) | Oxygen attacks S; C–O bond never breaks!

ROTs + Nu^- (SN2) | ROTs → R–Nu (INVERSION of R/S) | Overall 2-step sequence = net inversion

$SOCl_2$ or PBr_3 | ROH → R–Cl or R–Br (INVERSION) | Direct SN2 without carbocation rearrangement!

Sulfonate Ester Formation (TsCl or MsCl + Pyridine) — — Converts poor –OH into an excellent –OTs or –OMs leaving group with 100% RETENTION of configuration because the C–O bond stays intact.

Halide Conversion Without Rearrangement ($SOCl_2$ or PBr_3) — — Converts 1°/2° alcohols into R–Cl or R–Br with clean SN2 INVERSION—avoids the carbocation rearrangements seen with HCl/HBr!

Alcohol Protection (TMSCl / TBDMSCl or Dihydropyran) — — Masks acidic/nucleophilic –OH as a silyl ether (R–$OSiR_3$) during Grignard/$LiAlH_4$ steps; deprotect selectively with fluoride (TBAF).

activation:cue | Memory Cues

Activation changes what can leave. Protection temporarily changes what can react. Traditionally, an aldehyde-derived acetal carbon has H and R; a ketone-derived ketal carbon has two carbon groups. Both bear two OR groups. The broad IUPAC class acetals includes ketals.

Visual card: Stereochemical Control: Retention vs. Inversion from a Chiral Alcohol

How do you control the stereochemistry when replacing a chiral 2° –OH?

Need Net INVERSION from (R)-ROH — Route A: 1. TsCl/pyr (Retain) → 2. Nu^- (Invert) | Route B: PBr_3 (Invert) — One SN2 displacement at carbon gives net inversion: (R)-ROH → (R)-ROTs → (S)-R–Nu.

Need Net RETENTION from (R)-ROH — Double Inversion: 1. PBr_3 (Invert to S-RBr) → 2. Nu^- (Invert back to R-RNu) — Two sequential SN2 displacements at the chiral carbon invert twice (R → S → R) for net retention!

Grignard or Organolithium Fails on Halo-Alcohol — Protect –OH as R–OTBDMS First → Form RMgBr → Deprotect (F^-) — Unprotected –OH (pKa ~16) instantly protonates and destroys RMgBr (conjugate acid pKa ~50).

activation:trap | Common Mistakes / MCAT Traps

Making a tosylate does not itself perform SN2 at carbon. Acetals are generally acid-labile; acid and water are needed for hydrolysis. Protection does not make a group universally inert.

Visual card: Inverting the Stereocenter During Tosylation (ROH + TsCl)

Common Trap — — Flipping the wedge to a dash when converting (R)-butan-2-ol to its tosylate ester with TsCl/pyridine.

Correct Reasoning — — Tosylation happens at the OXYGEN atom (O attacks S of TsCl); the chiral C–O bond never breaks during tosylation, so configuration is RETAINED!

activation:app | MCAT Applications

A carbonyl can be protected while another group reacts, then regenerated. Choose protection and deprotection conditions compatible with the rest of the molecule.

Visual card: Biological Sulfonation, Kinase Phosphorylation & Solid-Phase Synthesis

ATP Phosphorylation of –OH — Biochemical Equivalent of Tosylation — Converting R–OH to R–OPO_3^{2-} turns a terrible leaving group (HO^-) into a weak-base phosphate leaving group ($HOPO_3^{2-}$).

Fluoride Deprotection (TBAF) — Si–F Bond Is Exceptionally Strong — Fluoride (F^-) selectively cleaves silyl ethers (R–OTBDMS) without touching carbon-oxygen ether or ester bonds.

HX vs PBr_3 on Neopentyl-like 2° ROH — HBr Gives Rearranged 3° RBr; PBr_3 Gives Clean 2° RBr — Use PBr_3 or TsCl/Br^- whenever you must prevent a 1,2-hydride or methyl shift.

Screenshot: activation-visuals



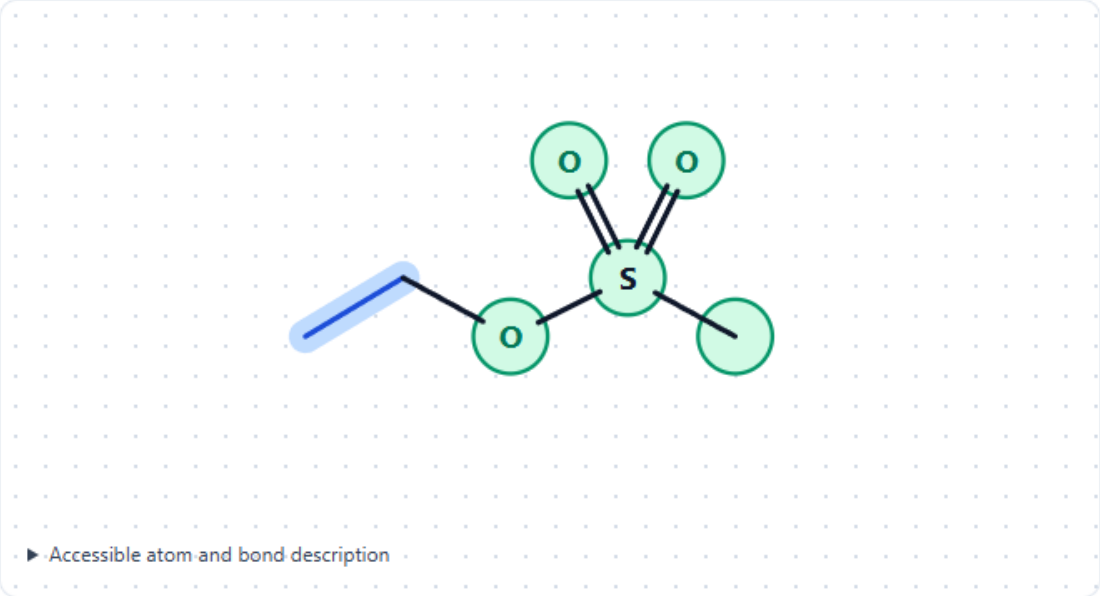
activation:step-1 | 1. Sulfonate ester

The ethyl group retains its C–O bond. The oxygen is now part of a sulfonate ester, whose leaving anion is resonance stabilized.

Displayed molecule ID: mesylate

{"highlights": {"prin": [2, 3, 4, 5, 6], "path": [0, 1]}, "revealName": true}

Screenshot: activation-step-1



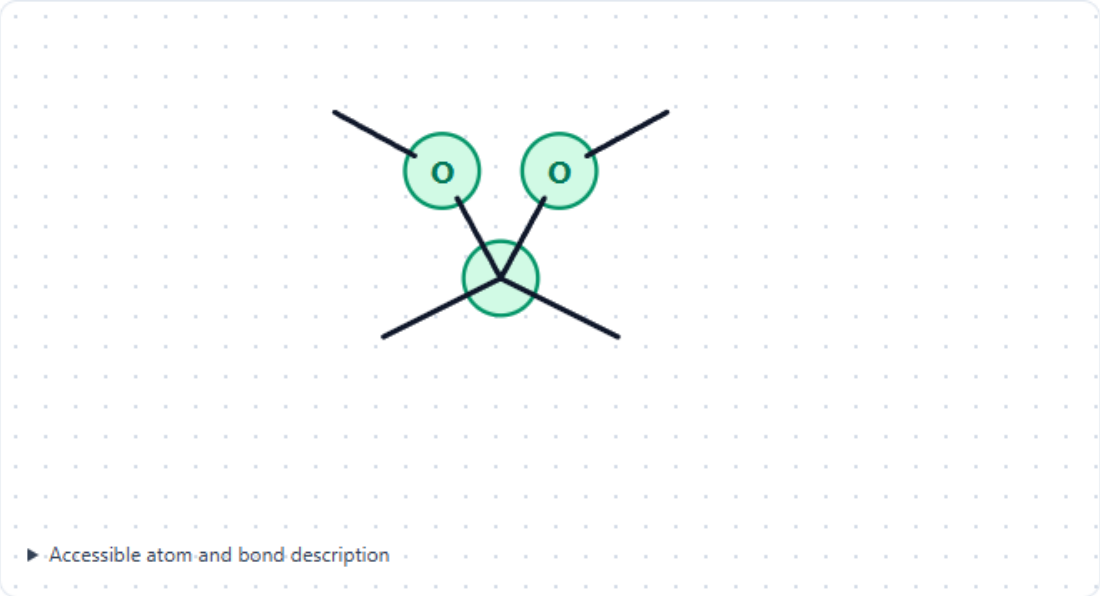
activation:step-2 | 2. Ketal protection

This ketal (a ketone-derived acetal) has two OR groups and two carbon groups at its central carbon. Acid-catalyzed reaction of a carbonyl with alcohol can form acetals or ketals; removing water favors formation.

Displayed molecule ID: acetal

{"highlights": {"prin": [1, 3, 4]}, "revealName": true}

Screenshot: activation-step-2



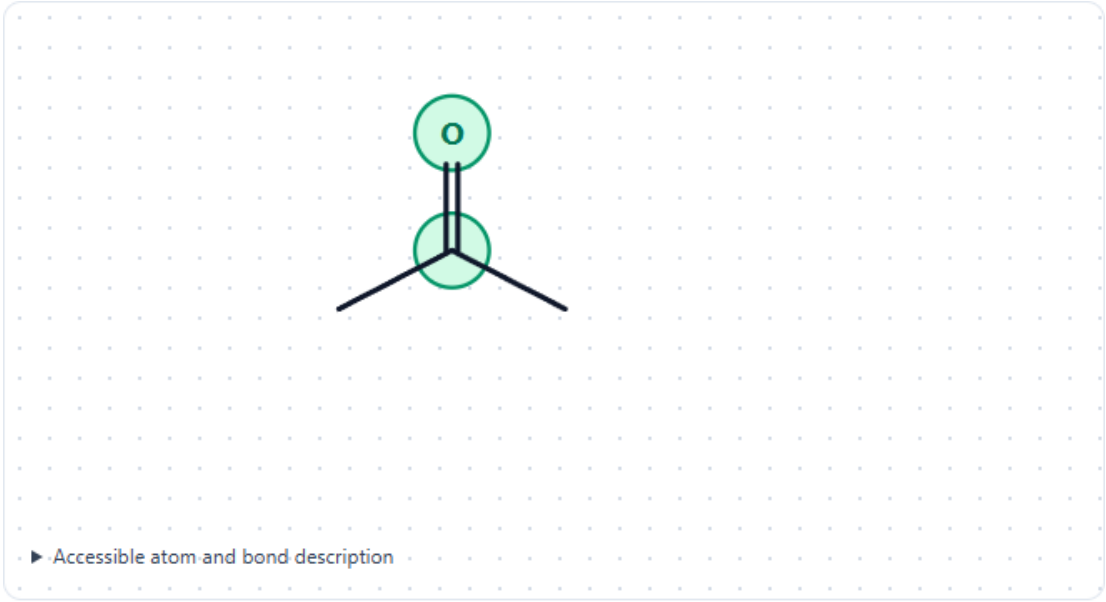
activation:step-3 | 3. Restore the carbonyl

Aqueous acid hydrolyzes this ketal, regenerating the ketone and alcohol. Protection and deprotection depend on the water/alcohol balance and compatible reaction conditions.

Displayed molecule ID: acetone

{"highlights": {"prin": [1, 3]}, "revealName": true}

Screenshot: activation-step-3



phenols | Phenols, quinones, and hydroquinones

Acidity and biological electron transfer

phenols:rule | Core Rules

Phenol is more acidic than an ordinary aliphatic alcohol because phenoxide can delocalize negative charge into the ring. Electron-withdrawing substituents can increase acidity. Hydroquinone can be oxidized to a quinone; the reverse is reduction.

Visual card: Phenol Acidity (pKa ~10), Substituent Resonance Effects & Hydroquinone ⇌ Quinone Redox

Chemistry diagram: Oxidation removes 2 H+ and 2 electrons; reverse reduction adds them. Schematic redox pair.

Hydroquinone (Aromatic 1,4-Benzenediol) ⇌ p-Benzoquinone (Conjugated Diketone, NOT Aromatic) + 2 H^+ + 2 e^-

Phenol pKa ~10 (vs ROH ~16)

Phenoxide Resonance = 4 Contributors

EWG (p-NO_2) → ↓ pKa (~7.1)

Hydroquinone ⇌ Quinone + $2H^+$ + $2e^-$

Cyclohexanol (pKa 16) | Localized sp^3 alkoxide | Insoluble in aqueous NaOH

Phenol (pKa 10.0) | Resonance-delocalized phenoxide | Dissolves in 1M NaOH (NOT in $NaHCO_3$!)

p-Nitrophenol (pKa 7.1) | Ortho/para EWG stabilizes O^- | Stronger acid by ~3 pKa units

Why Phenol (pKa ~10) Is 10^6× More Acidic Than Cyclohexanol (pKa ~16) — — Deprotonating phenol delocalizes the negative charge across oxygen AND the ortho/para ring carbons (4 resonance contributors).

Why Phenol Is Less Acidic Than Carboxylic Acids (pKa ~5) — — Carboxylate spreads −1 charge over TWO electronegative oxygens; phenoxide spreads charge onto less electronegative carbon atoms.

phenols:cue | Memory Cues

Quinone + 2 H+ + 2 e− → hydroquinone. “Hydro” here tracks reduction, not merely adding OH groups.

Visual card: Phenol Substituent Effects & Quinone Redox Recognition

What phenol or quinone property is being tested?

Nitro (–NO_2) or Cyano (–CN) on Phenol Ring — ↓↓ pKa (Stronger Acid — Especially Ortho/Para!) — Ortho and para EWGs accept the phenoxide lone pair directly by resonance + induction (2,4,6-trinitrophenol = picric acid, pKa ~0.4!).

Methoxy (–OCH_3) or Alkyl (–CH_3) on Phenol Ring — ↑ pKa (Weaker Acid) — Electron-donating groups push electron density into the ring, destabilizing the negatively charged phenoxide conjugate base.

Hydroquinone / Catechol + Oxidant ([O]) — Oxidizes to Conjugated Quinone (1,4 or 1,2-Dione) — Quinones are cyclic α,β-unsaturated diketones (NOT aromatic!), whereas hydroquinones ARE fully aromatic 6π benzene diols.

phenols:trap | Common Mistakes / MCAT Traps

Hydroquinone and hydroxyquinone are different terms. Hydroquinone is a reduced quinone form; a hydroxyquinone still has a quinone framework bearing an extra OH. Coenzyme Q accepts electrons upstream of complex III and delivers them to complex III. Oxidative hydroxylation to a hydroxyquinone is condition-dependent; it is not a universal second oxidation step after quinone formation.

Visual card: Calling Quinones "Aromatic" or Extracting Phenol with $NaHCO_3$

Common Trap — — Assuming ubiquinone (CoQ) is aromatic because it has a 6-membered ring, or saying $NaHCO_3$ deprotonates unsubstituted phenol.

Correct Reasoning — — Benzoquinone has two sp^2 C=O carbons interrupting cyclic delocalization (non-aromatic conjugated diene-dione); phenol (pKa 10) needs NaOH (H_2O pKa 15.7), not $NaHCO_3$ (pKa 6.4)!

phenols:app | MCAT Applications

Ubiquinone/ubiquinol transports electrons within the mitochondrial inner membrane; proton-coupled redox chemistry links the organic structure to energy metabolism. CoQ accepts electrons from complexes I and II and passes them to III. Its isoprenoid tail keeps it in the membrane. Vitamin K1 (phylloquinone) and K2 (menaquinones) are lipid-soluble quinones with hydrophobic side chains and roles in vitamin K-dependent carboxylation.

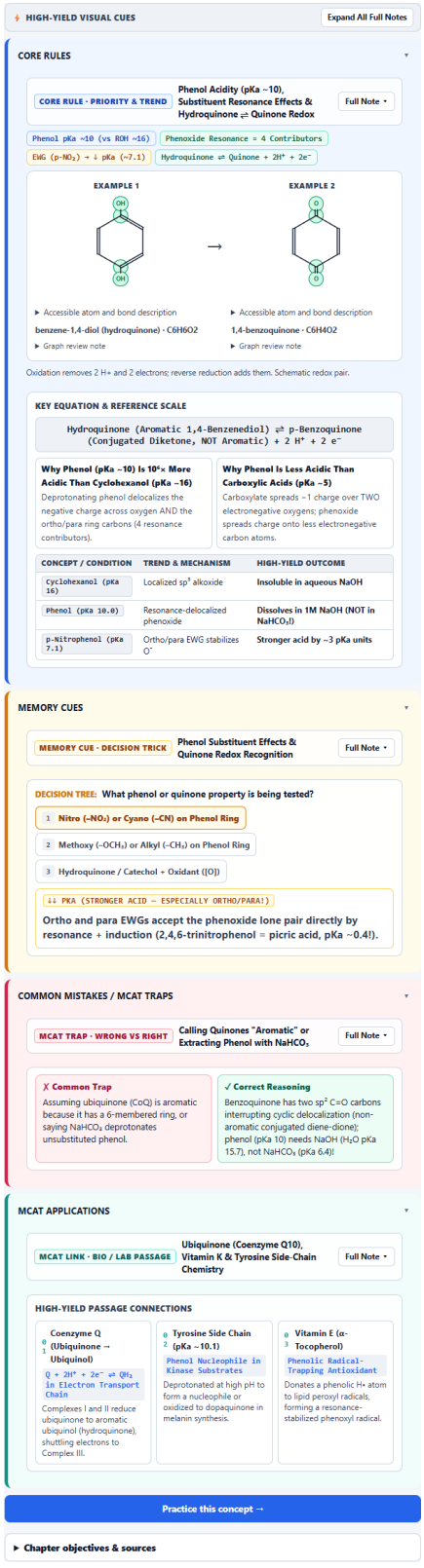
Visual card: Ubiquinone (Coenzyme Q10), Vitamin K & Tyrosine Side-Chain Chemistry

Coenzyme Q (Ubiquinone → Ubiquinol) — $Q + 2H^+ + 2e^- \rightleftharpoons QH_2$ in Electron Transport Chain — Complexes I and II reduce ubiquinone to aromatic ubiquinol (hydroquinone), shuttling electrons to Complex III.

Tyrosine Side Chain (pKa ~10.1) — Phenol Nucleophile in Kinase Substrates — Deprotonated at high pH to form a nucleophile or oxidized to dopaquinone in melanin synthesis.

Vitamin E (α-Tocopherol) — Phenolic Radical-Trapping Antioxidant — Donates a phenolic H• atom to lipid peroxyl radicals, forming a resonance-stabilized phenoxyl radical.

Screenshot: phenols-visuals



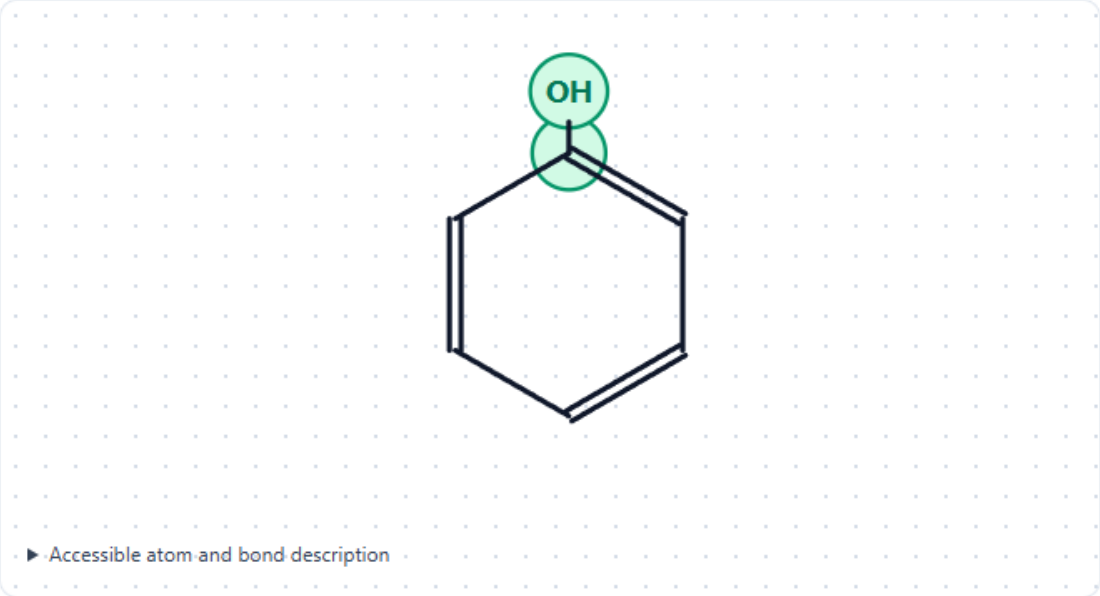
phenols:step-1 | 1. Phenolic OH

OH is attached directly to an aromatic carbon. Deprotonation gives phenoxide, stabilized by resonance. The pictured alternating bonds are one aromatic resonance representation.

Displayed molecule ID: phenol

{"highlights": {"prin": [0, 6]}, "revealName": true}

Screenshot: phenols-step-1



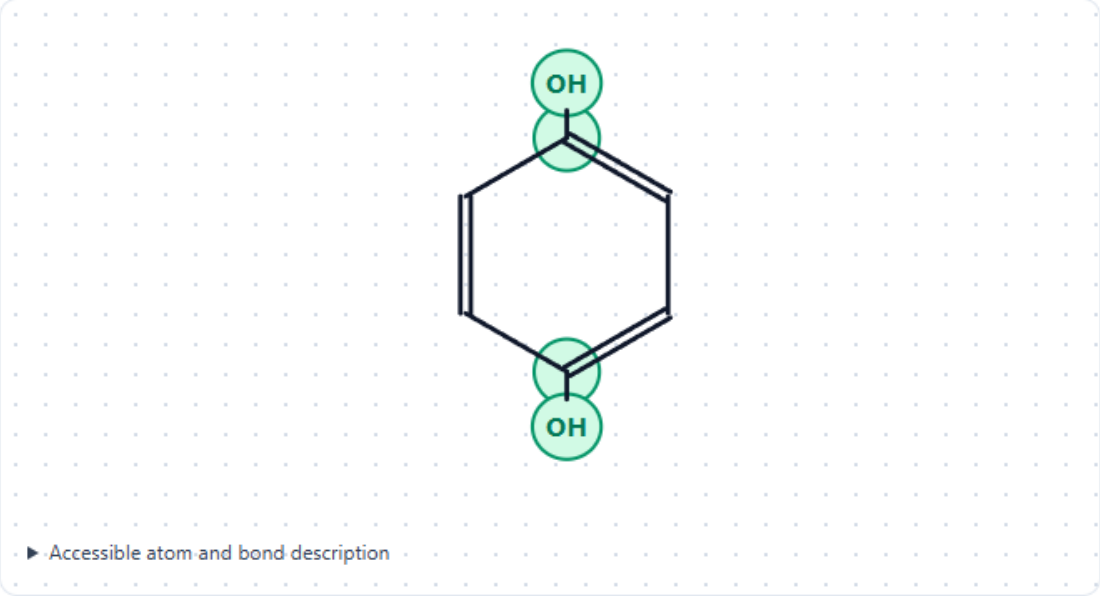
phenols:step-2 | 2. Reduced member

Hydroquinone has two phenolic OH groups in a para arrangement. Oxidation removes two H atoms overall, formally two protons plus two electrons.

Displayed molecule ID: hydroquinone

{"highlights": {"prin": [0, 3, 6, 7]}, "revealName": true}

Screenshot: phenols-step-2



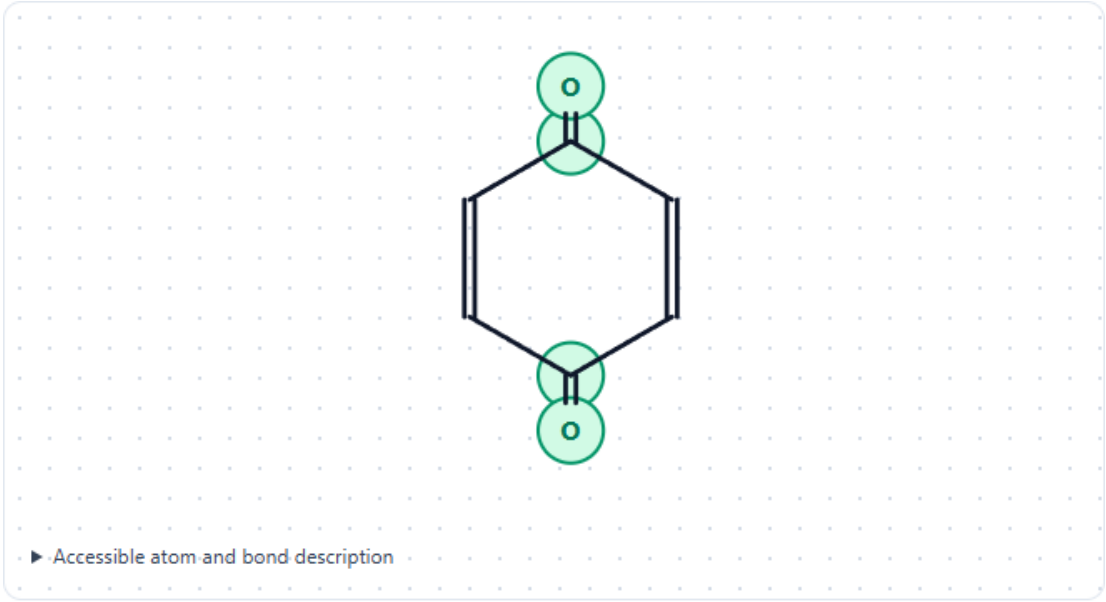
phenols:step-3 | 3. Oxidized member

The quinone has two carbonyls. Its ring is not simply benzene with unchanged aromatic bonding; the pi-bond arrangement has changed.

Displayed molecule ID: quinone

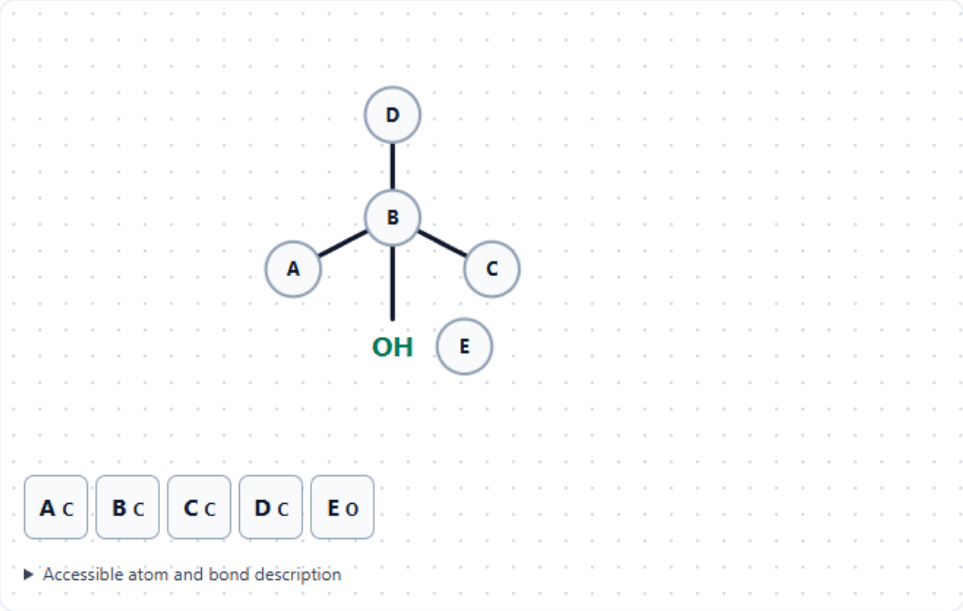
{"highlights": {"prin": [0, 3, 6, 7]}, "revealName": true}

Screenshot: phenols-step-3



classify-alcohol | Classify by carbon neighbors

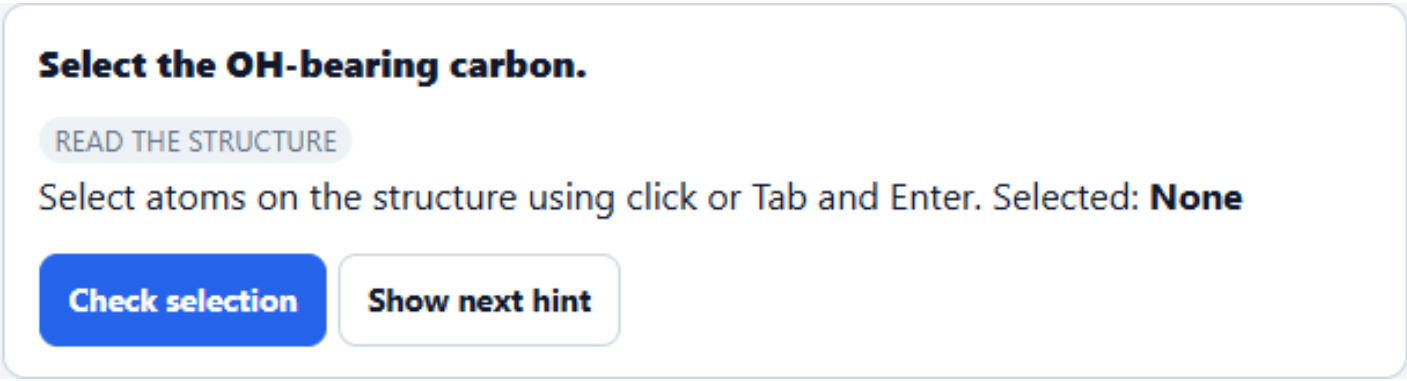
Screenshot: classify-alcohol-stage-1-molecule



classify-alcohol:stage-1 | Select the OH-bearing carbon.

Answer type: atoms

Screenshot: classify-alcohol-stage-1-question



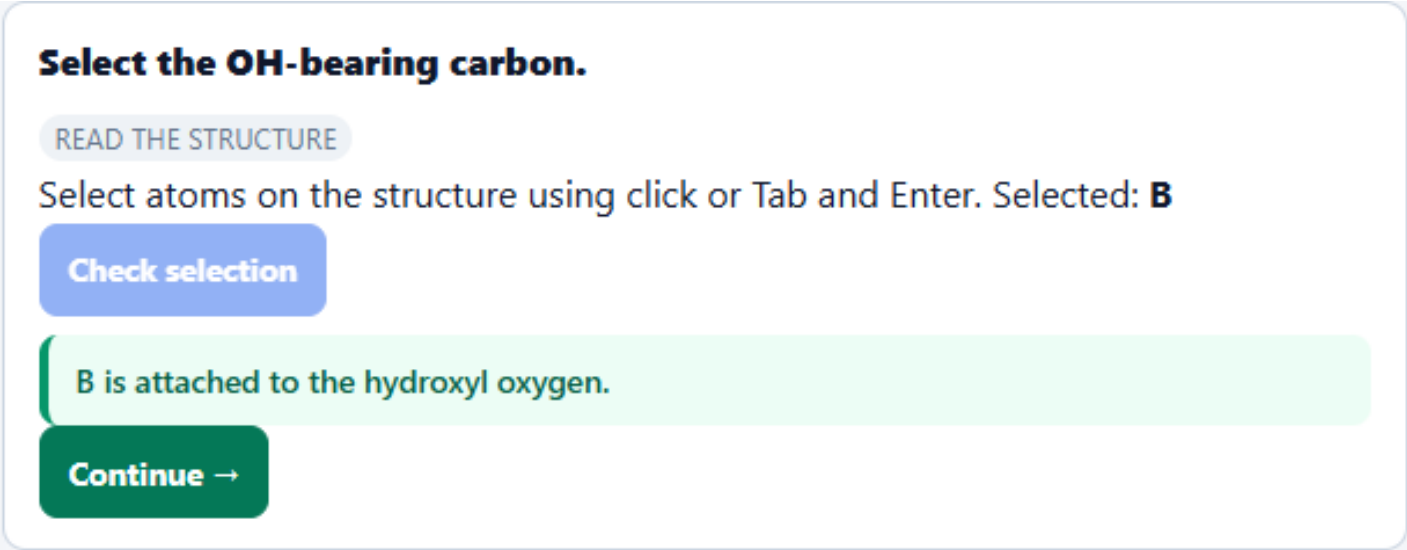
{"authorAnswer": [1], "tolerance": null, "highlightsAfterCorrectAnswer": {"prin": [1, 4]}}

Hint 1: Find the C–O bond.

Hint 2: Choose the carbon, not the oxygen.

Explanation: B is attached to the hydroxyl oxygen.

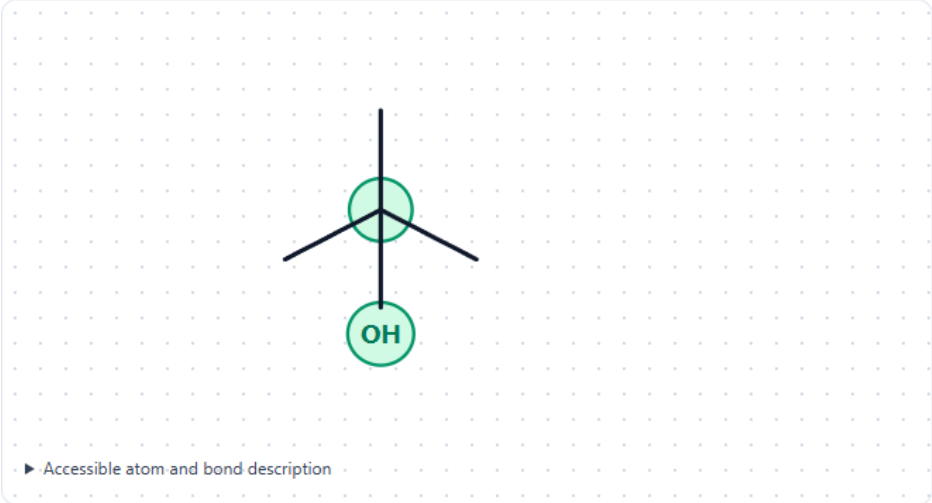
Screenshot: classify-alcohol-stage-1-feedback



classify-alcohol:stage-2 | How many carbon neighbors does it have?

Answer type: number | unit: carbon neighbors

Screenshot: classify-alcohol-stage-2-molecule



Screenshot: classify-alcohol-stage-2-question



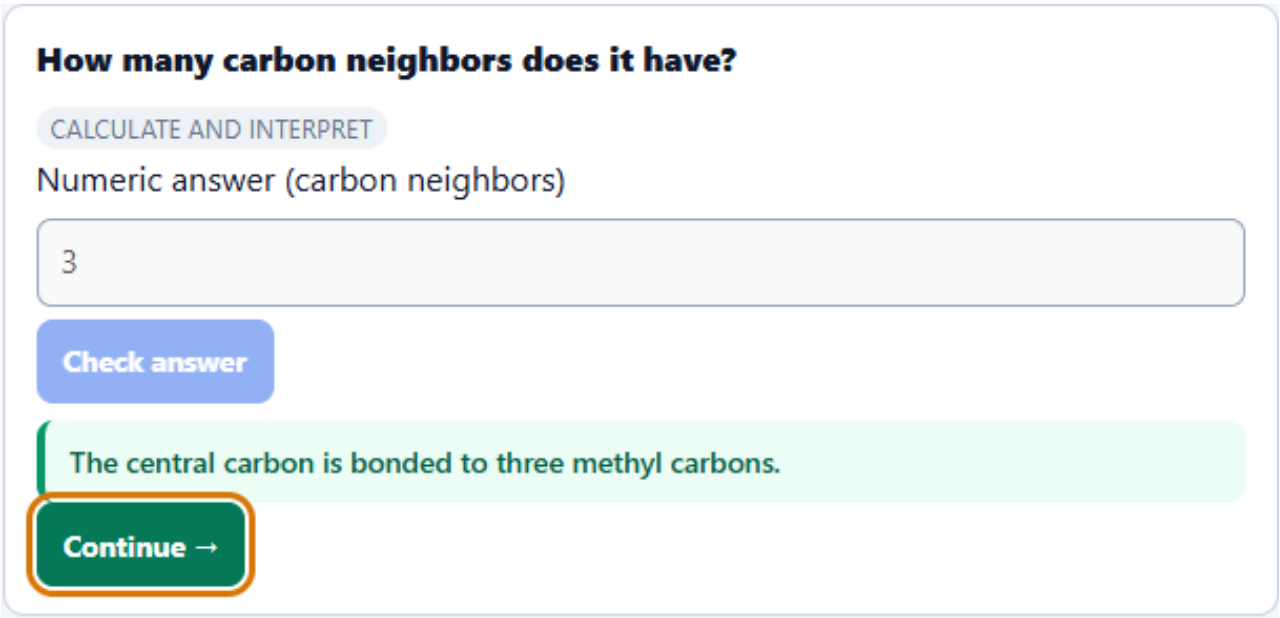
{"authorAnswer": 3, "tolerance": 0, "highlightsAfterCorrectAnswer": {}}

Hint 1: Count only directly attached C atoms.

Hint 2: Exclude oxygen and implicit H.

Explanation: The central carbon is bonded to three methyl carbons.

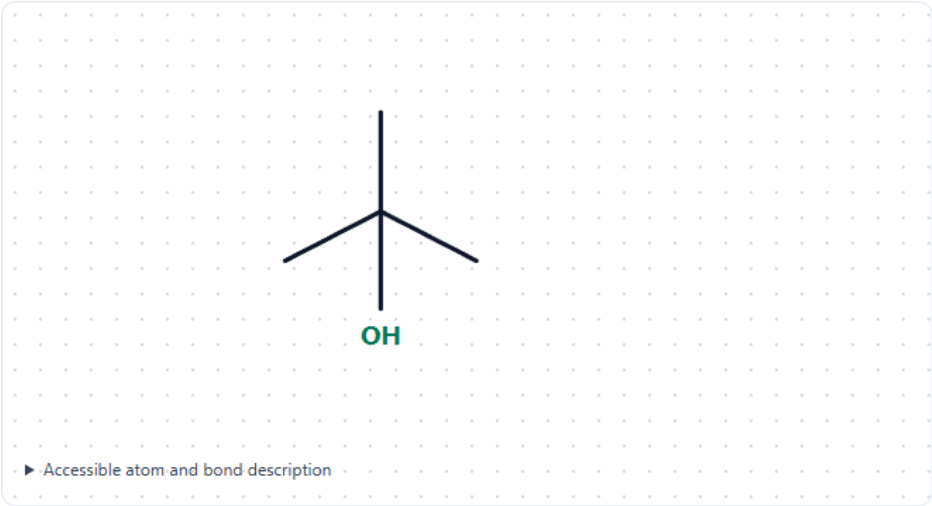
Screenshot: classify-alcohol-stage-2-feedback



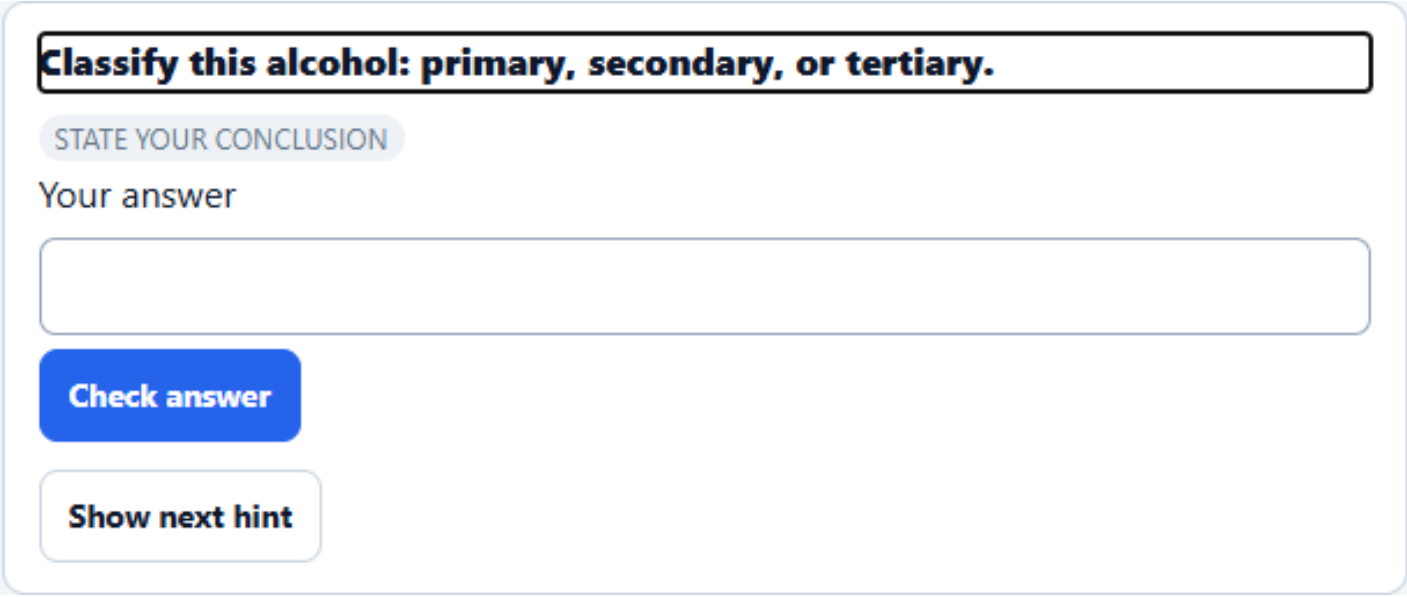
classify-alcohol:stage-3 | Classify this alcohol: primary, secondary, or tertiary.

Answer type: text

Screenshot: classify-alcohol-stage-3-molecule



Screenshot: classify-alcohol-stage-3-question



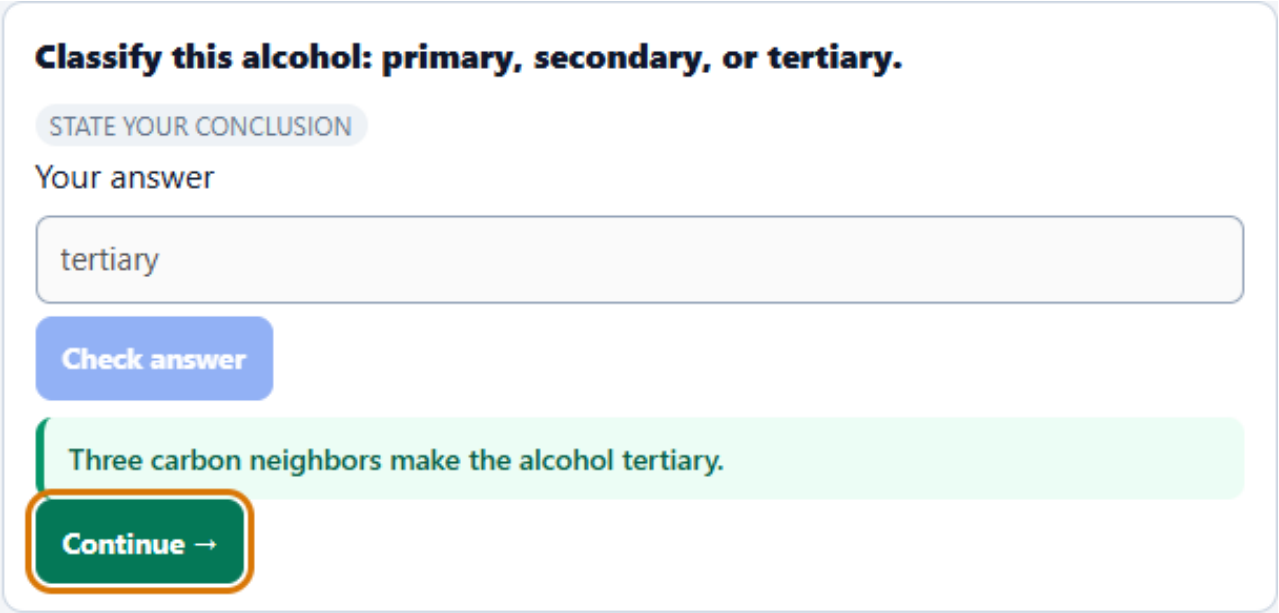
{"authorAnswer": ["tertiary", "3°", "3", "tertiary alcohol", "3° alcohol"], "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Use the carbon-neighbor count.

Hint 2: The OH-bearing carbon has no hydrogen.

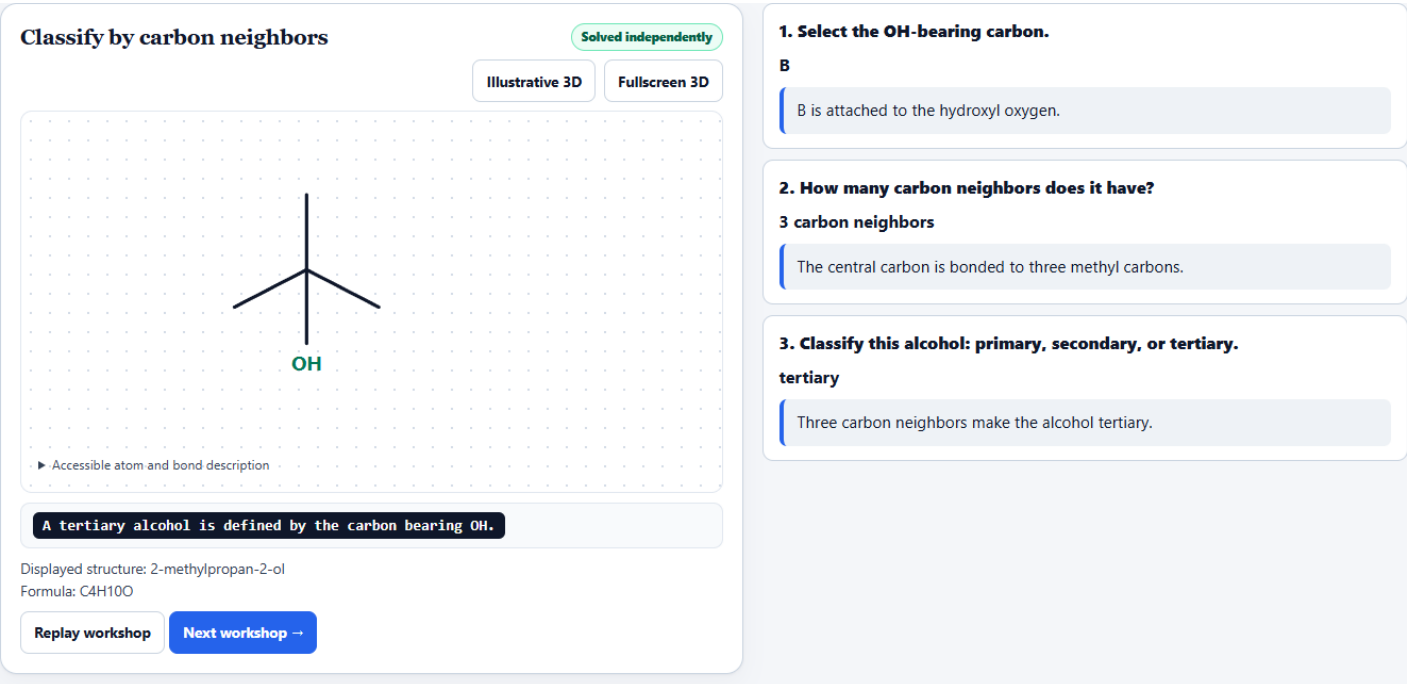
Explanation: Three carbon neighbors make the alcohol tertiary.

Screenshot: classify-alcohol-stage-3-feedback



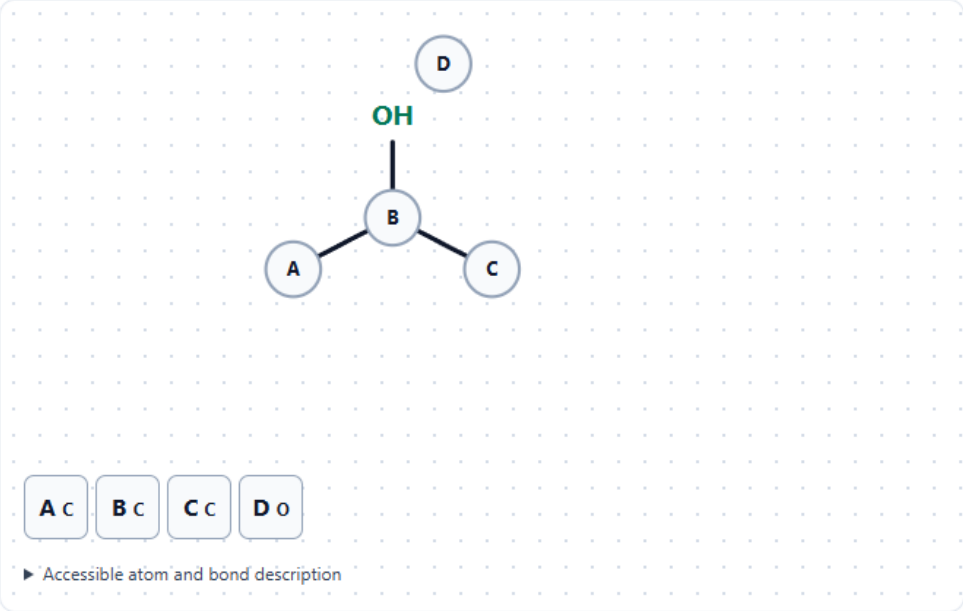
Completion summary: A tertiary alcohol is defined by the carbon bearing OH.

Screenshot: classify-alcohol-complete



oxidize-secondary | Predict the oxidation product

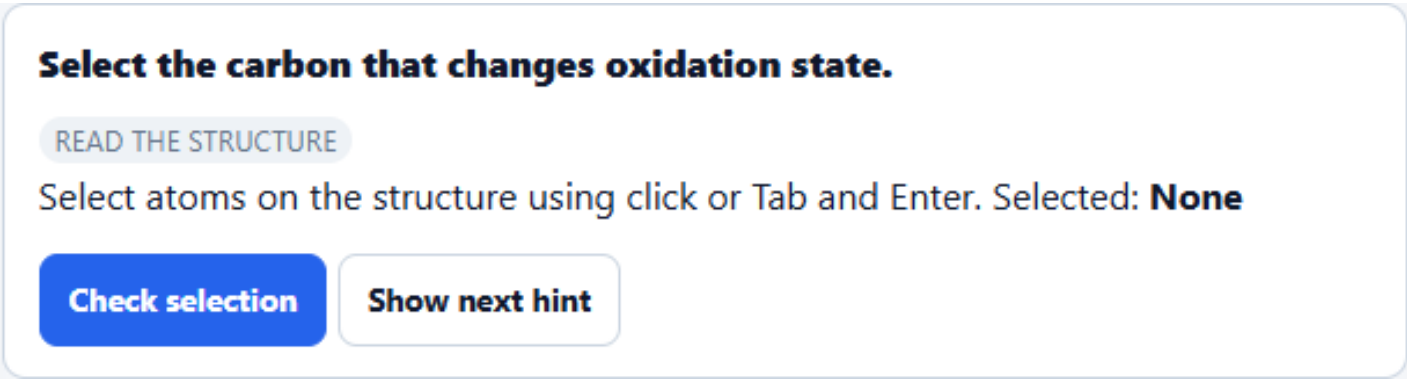
Screenshot: oxidize-secondary-stage-1-molecule



oxidize-secondary:stage-1 | Select the carbon that changes oxidation state.

Answer type: atoms

Screenshot: oxidize-secondary-stage-1-question



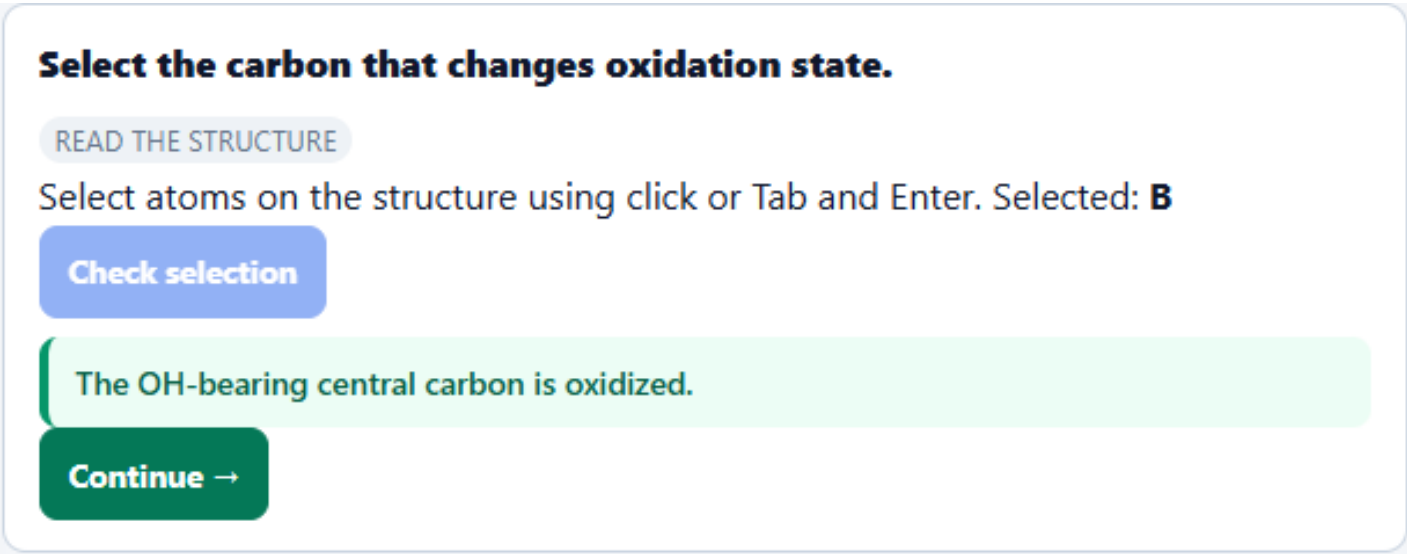
{"authorAnswer": [1], "tolerance": null, "highlightsAfterCorrectAnswer": {"prin": [1, 3]}}

Hint 1: The carbon skeleton stays intact.

Hint 2: Track the C–O bond.

Explanation: The OH-bearing central carbon is oxidized.

Screenshot: oxidize-secondary-stage-1-feedback



oxidize-secondary:stage-2 | Ordinary oxidation produces which functional group?

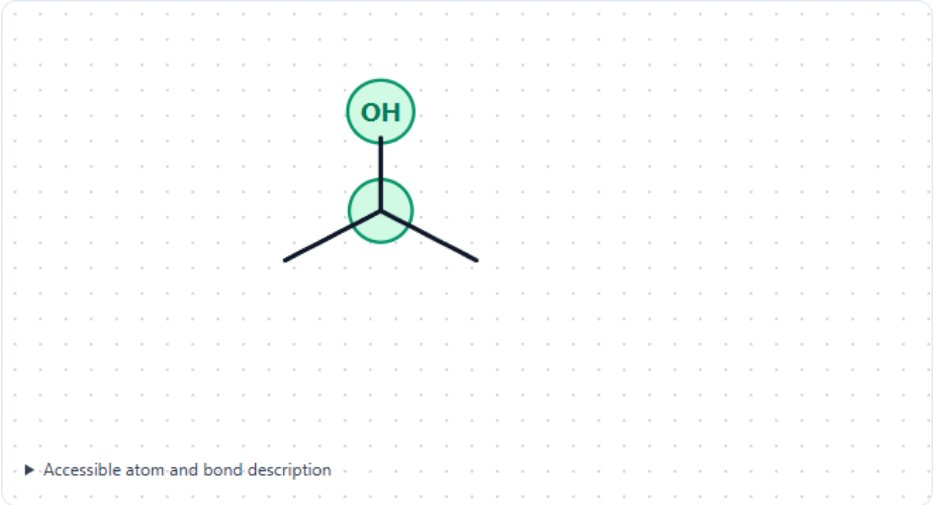
Answer type: choice

Option 0 (A): Ketone

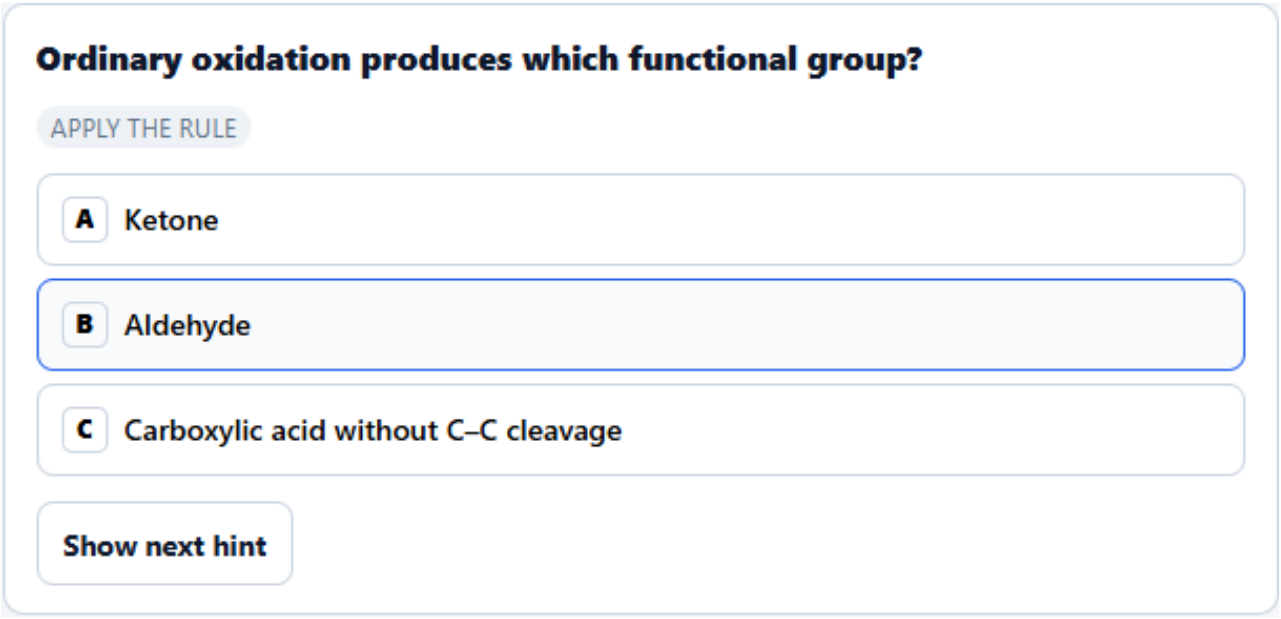
Option 1 (B): Aldehyde

Option 2 (C): Carboxylic acid without C–C cleavage

Screenshot: oxidize-secondary-stage-2-molecule



Screenshot: oxidize-secondary-stage-2-question



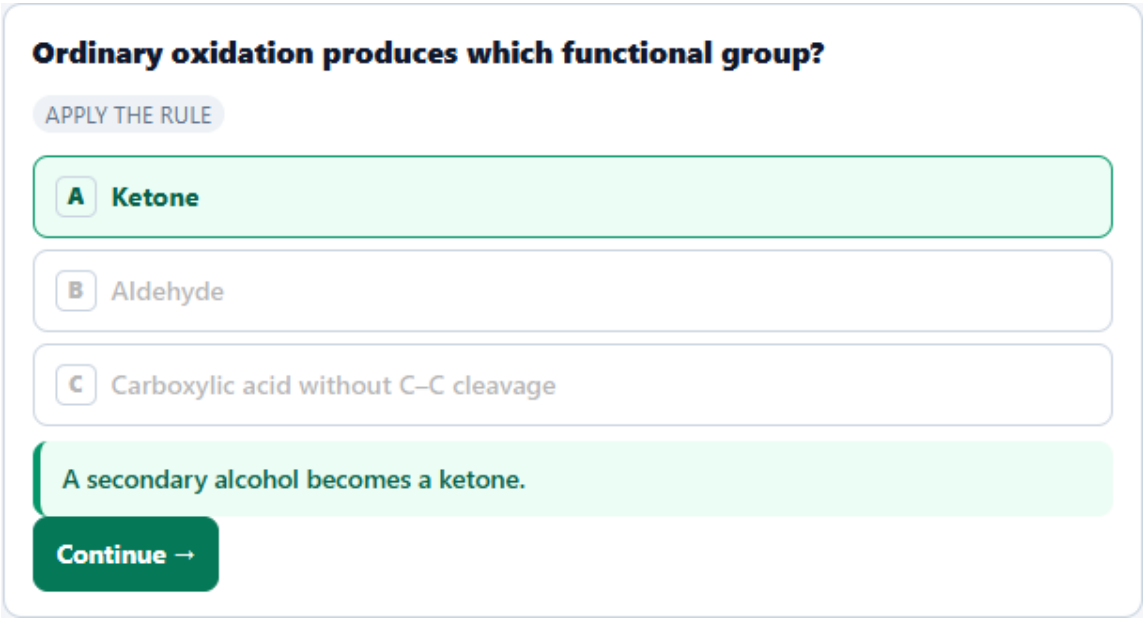
{"authorAnswer": 0, "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: How many carbon substituents remain on the carbonyl?

Hint 2: Two carbon neighbors indicate a ketone.

Explanation: A secondary alcohol becomes a ketone.

Screenshot: oxidize-secondary-stage-2-feedback

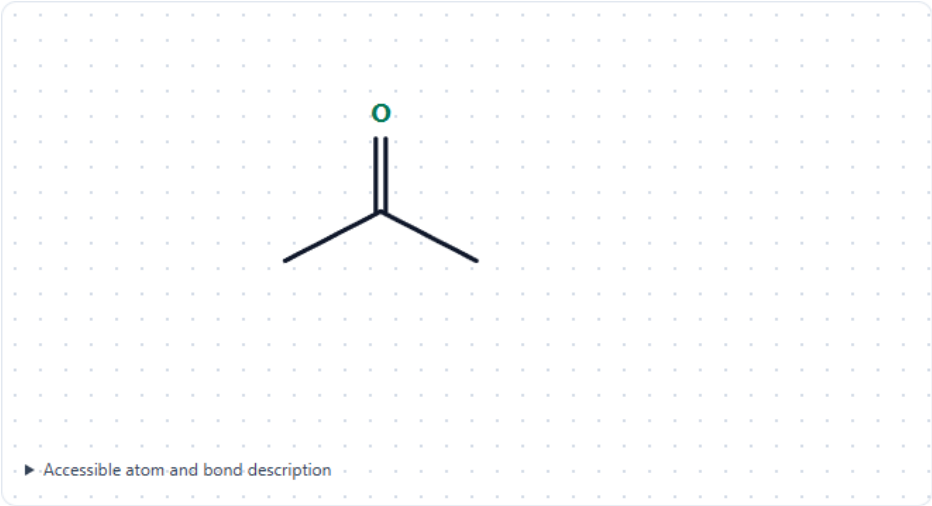


This stage uses a different molecule (connectivity shown).

oxidize-secondary:stage-3 | Name the three-carbon product.

Answer type: text

Screenshot: oxidize-secondary-stage-3-molecule



Screenshot: oxidize-secondary-stage-3-question



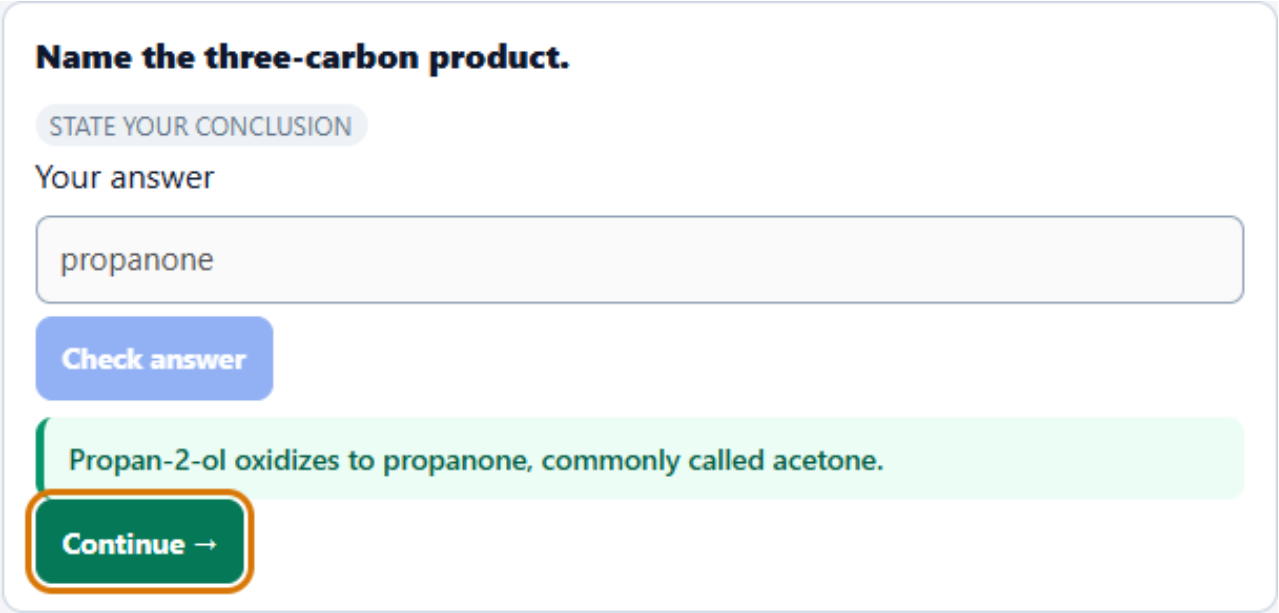
{"authorAnswer": ["propanone", "propan-2-one", "acetone", "2-propanone", "dimethyl ketone"], "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Retain all three carbons.

Hint 2: Replace the secondary alcohol with a ketone.

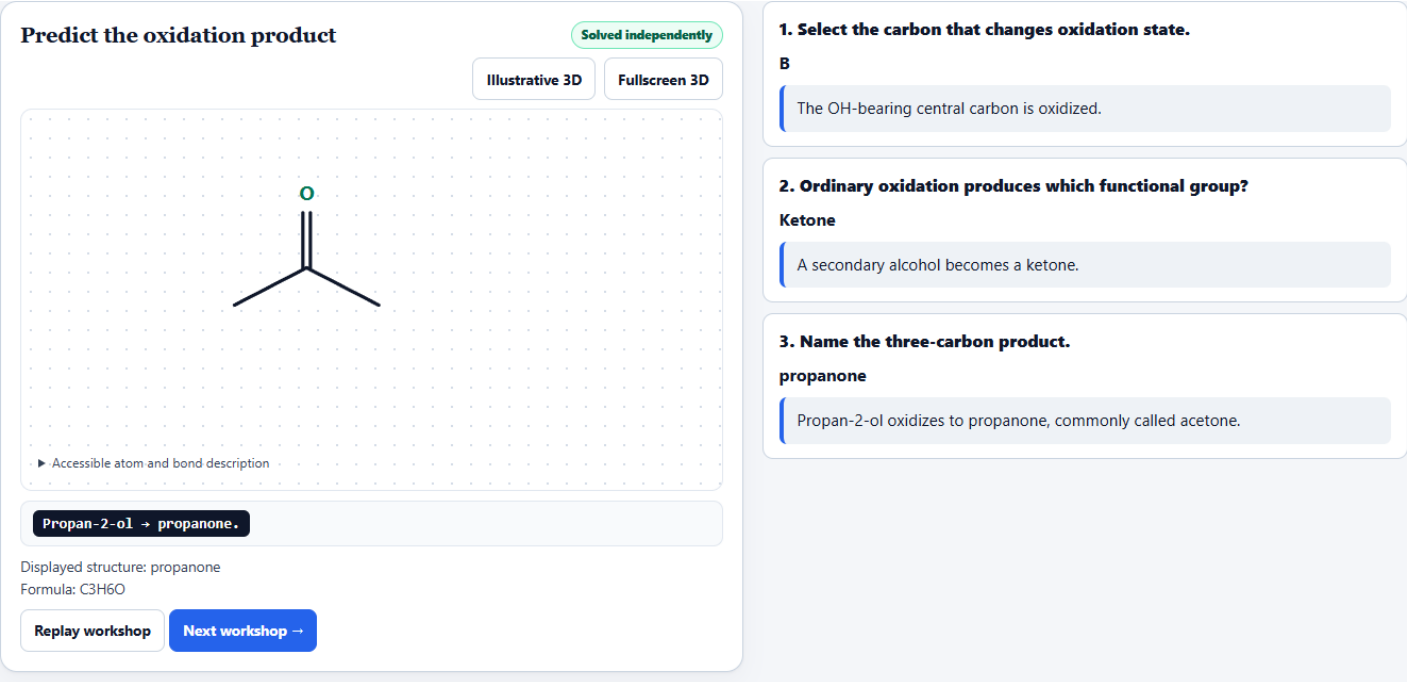
Explanation: Propan-2-ol oxidizes to propanone, commonly called acetone.

Screenshot: oxidize-secondary-stage-3-feedback



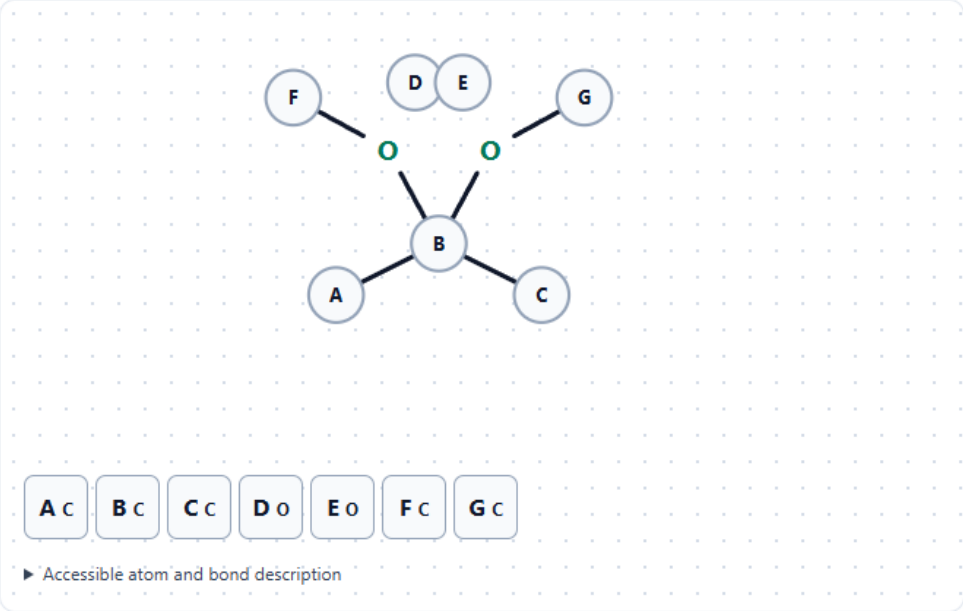
Completion summary: Propan-2-ol → propanone.

Screenshot: oxidize-secondary-complete



protection | Use a protecting group correctly

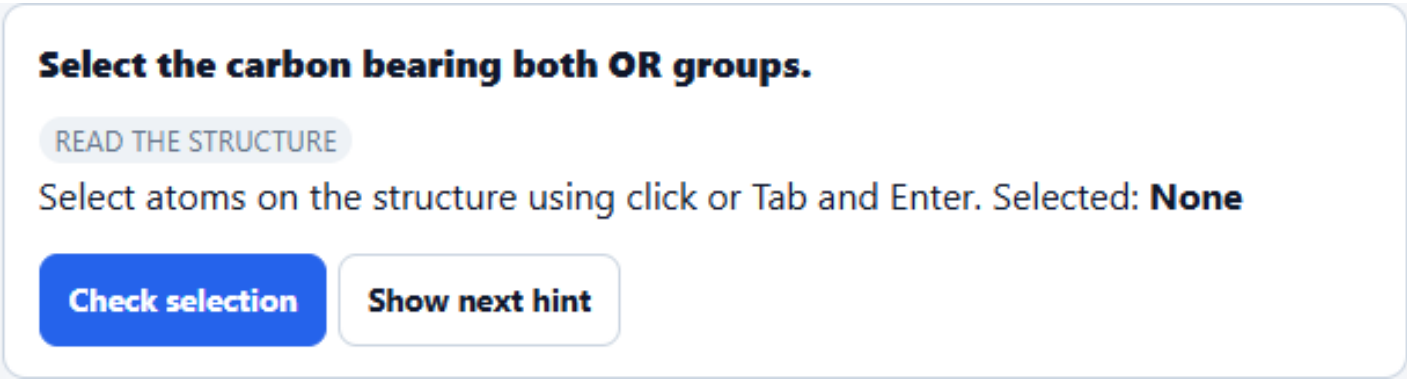
Screenshot: protection-stage-1-molecule



protection:stage-1 | Select the carbon bearing both OR groups.

Answer type: atoms

Screenshot: protection-stage-1-question



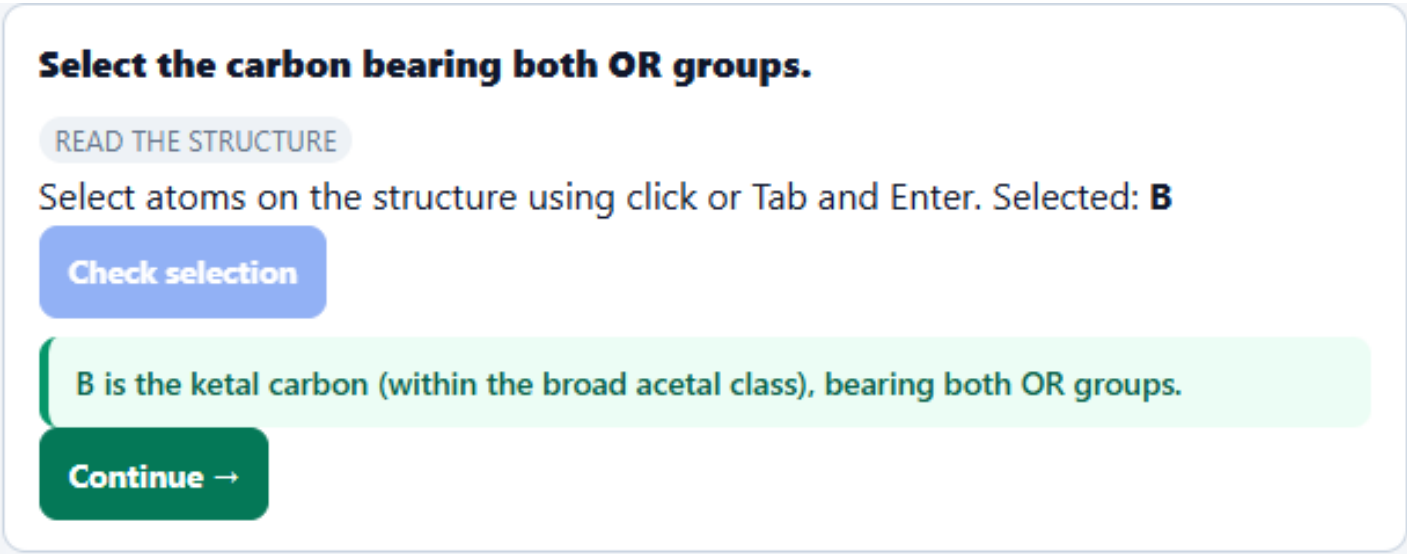
{"authorAnswer": [1], "tolerance": null, "highlightsAfterCorrectAnswer": {"prin": [1, 3, 4]}}

Hint 1: Find a carbon attached to two oxygens.

Hint 2: There is no C=O at that carbon.

Explanation: B is the ketal carbon (within the broad acetal class), bearing both OR groups.

Screenshot: protection-stage-1-feedback



protection:stage-2 | Which conditions ordinarily regenerate the ketone?

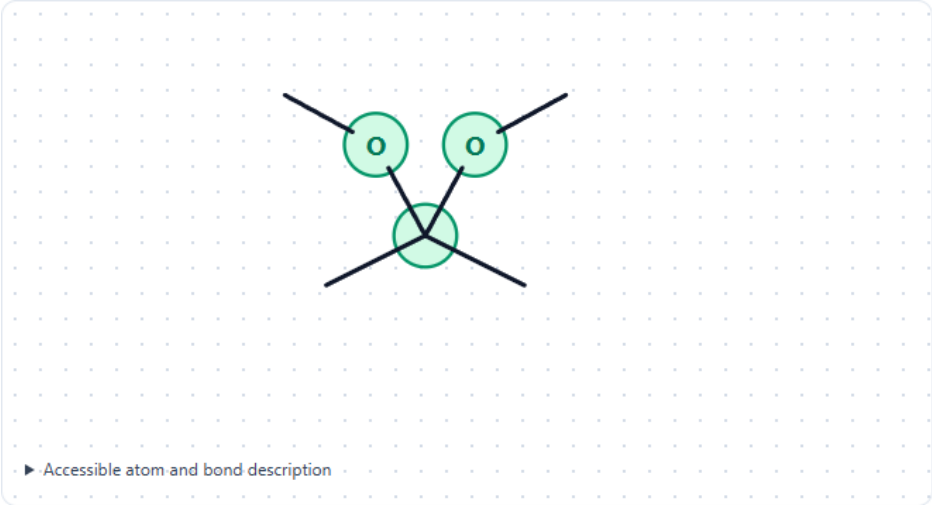
Answer type: choice

Option 0 (A): Aqueous acid

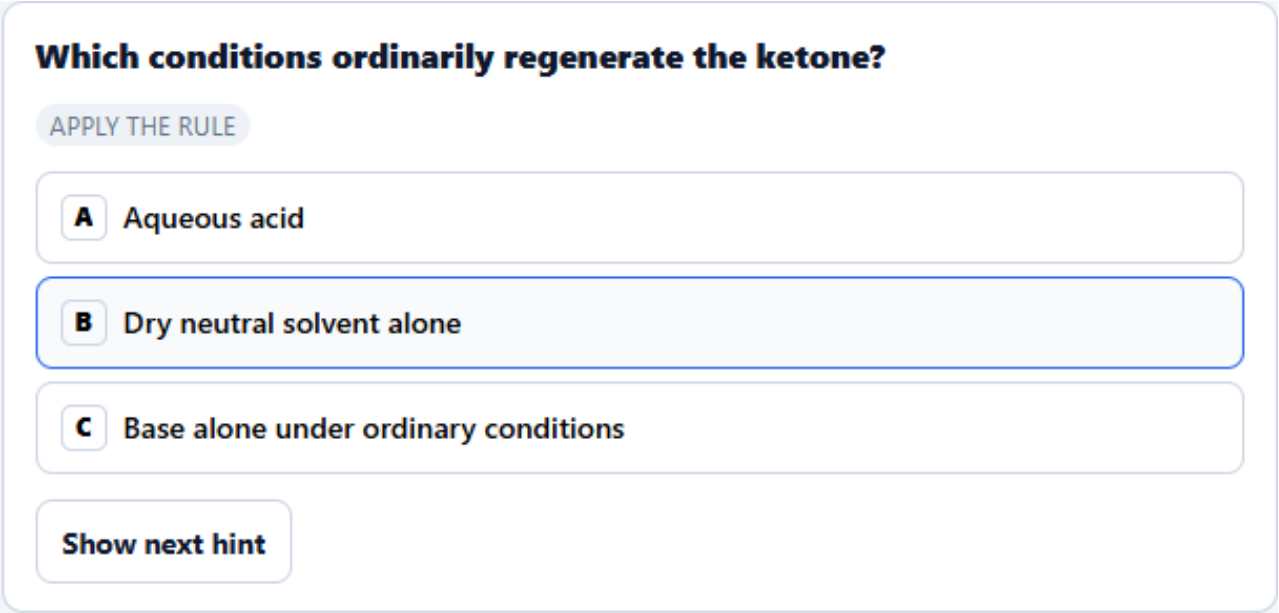
Option 1 (B): Dry neutral solvent alone

Option 2 (C): Base alone under ordinary conditions

Screenshot: protection-stage-2-molecule



Screenshot: protection-stage-2-question



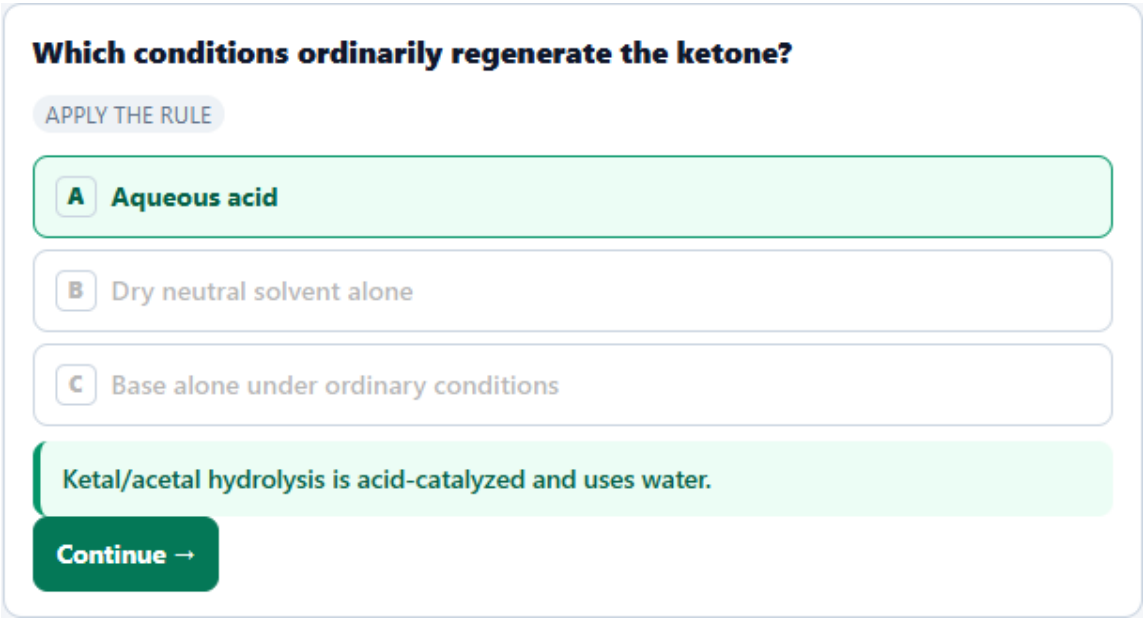
{"authorAnswer": 0, "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Consider the reverse of ketal/acetal formation.

Hint 2: Both catalyst and water are important.

Explanation: Ketal/acetal hydrolysis is acid-catalyzed and uses water.

Screenshot: protection-stage-2-feedback



This stage uses a different molecule (connectivity shown).

protection:stage-3 | Making an alkyl sulfonate from ROH (mesylate shown; tosylate analogous) directly causes…

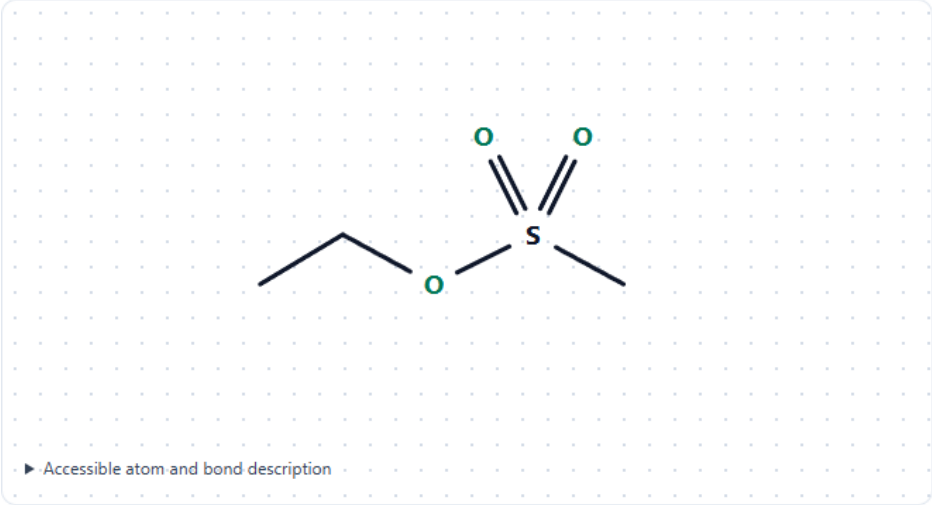
Answer type: choice

Option 0 (A): Retention of the C–O framework

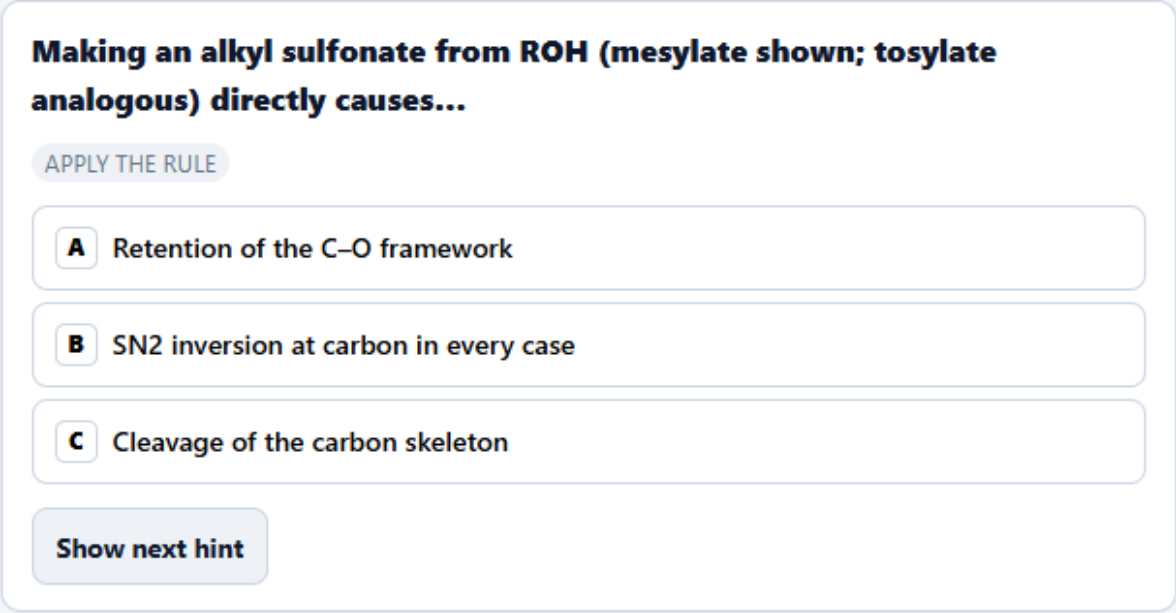
Option 1 (B): SN2 inversion at carbon in every case

Option 2 (C): Cleavage of the carbon skeleton

Screenshot: protection-stage-3-molecule



Screenshot: protection-stage-3-question



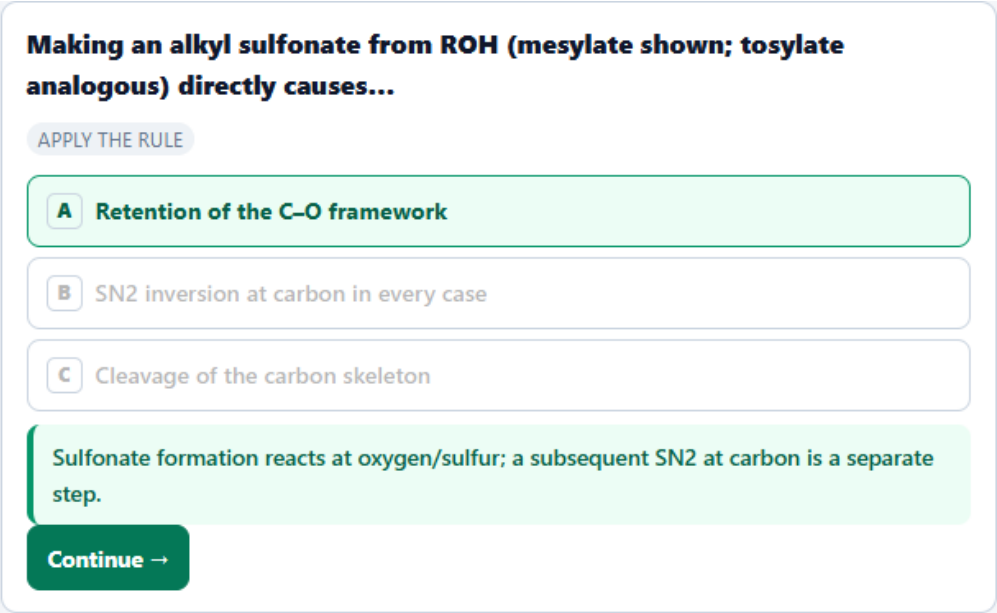
{"authorAnswer": 0, "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Distinguish activation from displacement.

Hint 2: The original C–O bond remains during activation.

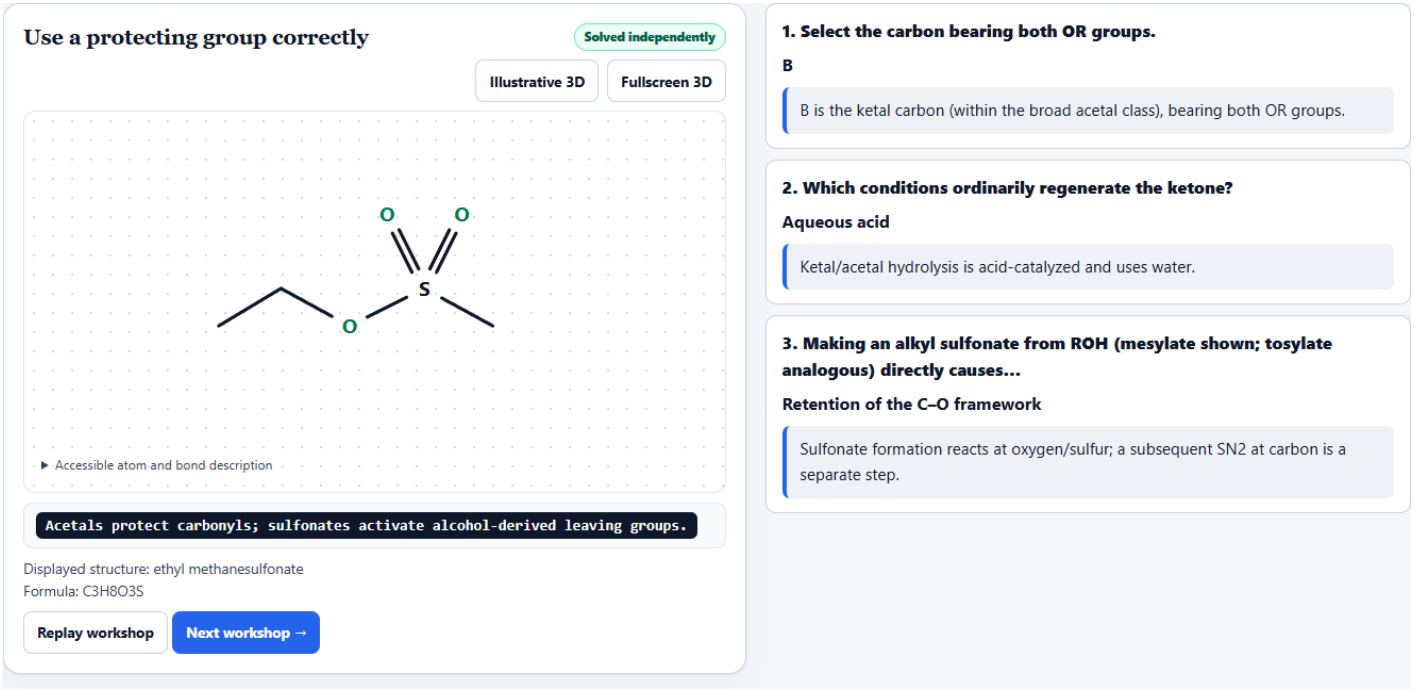
Explanation: Sulfonate formation reacts at oxygen/sulfur; a subsequent SN2 at carbon is a separate step.

Screenshot: protection-stage-3-feedback



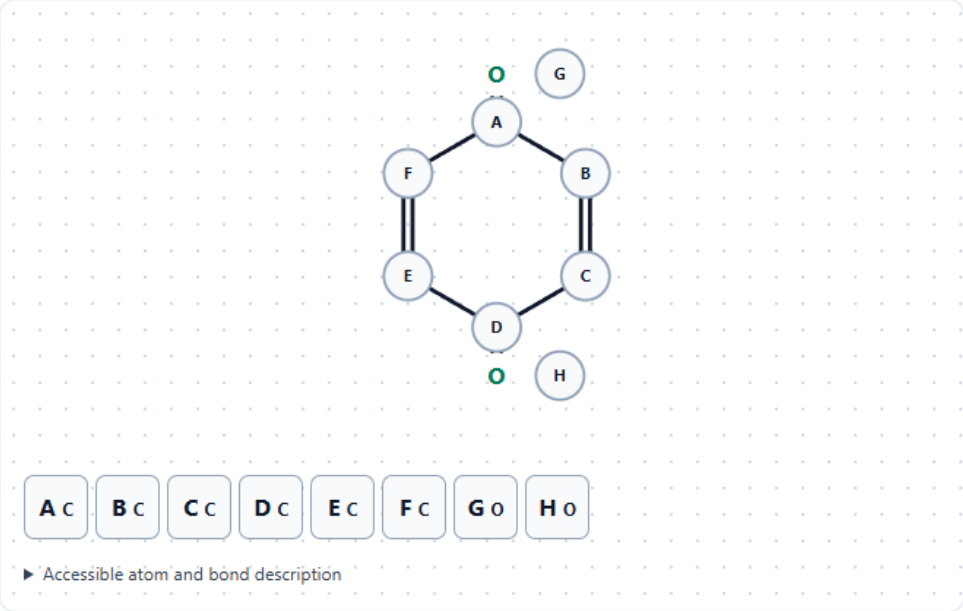
Completion summary: Acetals protect carbonyls; sulfonates activate alcohol-derived leaving groups.

Screenshot: protection-complete



quinone-redox | Track quinone reduction

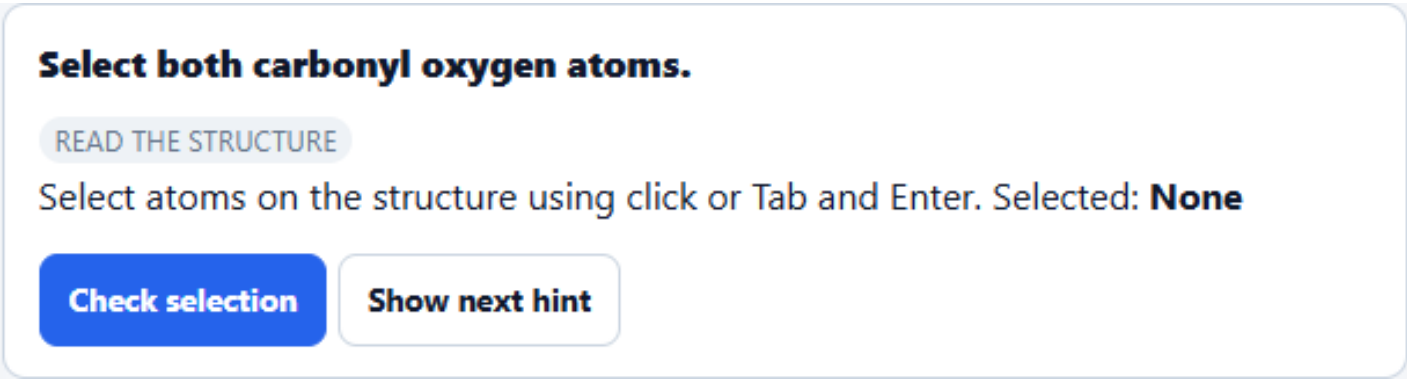
Screenshot: quinone-redox-stage-1-molecule



quinone-redox:stage-1 | Select both carbonyl oxygen atoms.

Answer type: atoms

Screenshot: quinone-redox-stage-1-question



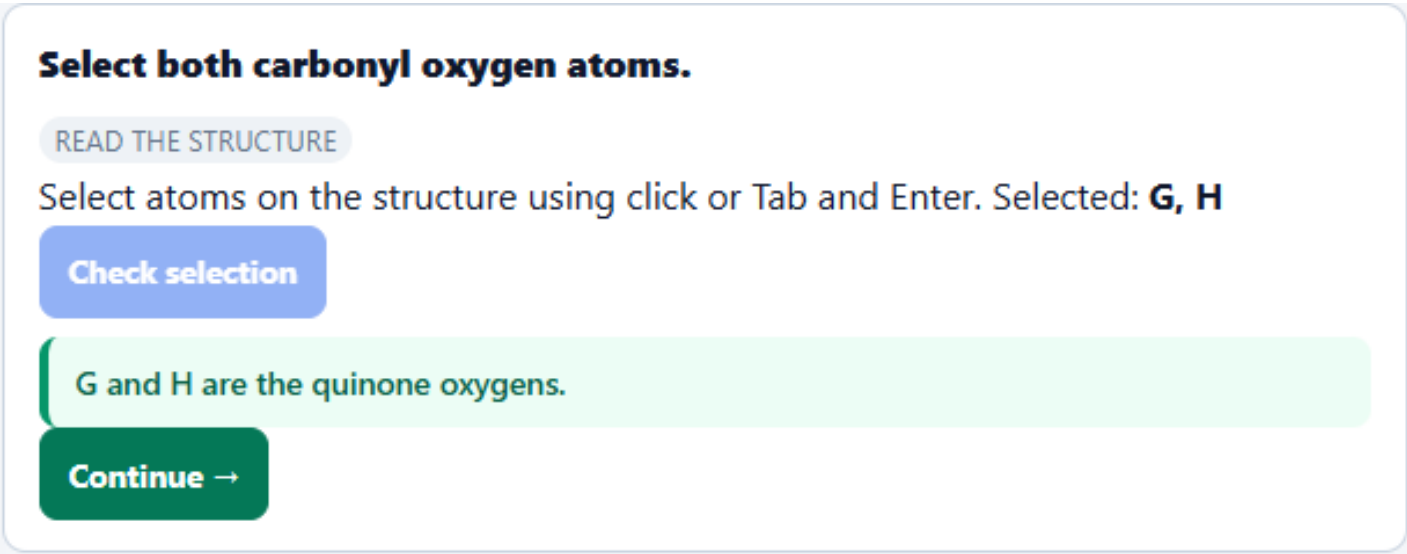
{"authorAnswer": [6, 7], "tolerance": null, "highlightsAfterCorrectAnswer": {"prin": [6, 7]}}

Hint 1: Identify the two C=O bonds.

Hint 2: Select the oxygen ends.

Explanation: G and H are the quinone oxygens.

Screenshot: quinone-redox-stage-1-feedback



quinone-redox:stage-2 | Quinone → hydroquinone is…

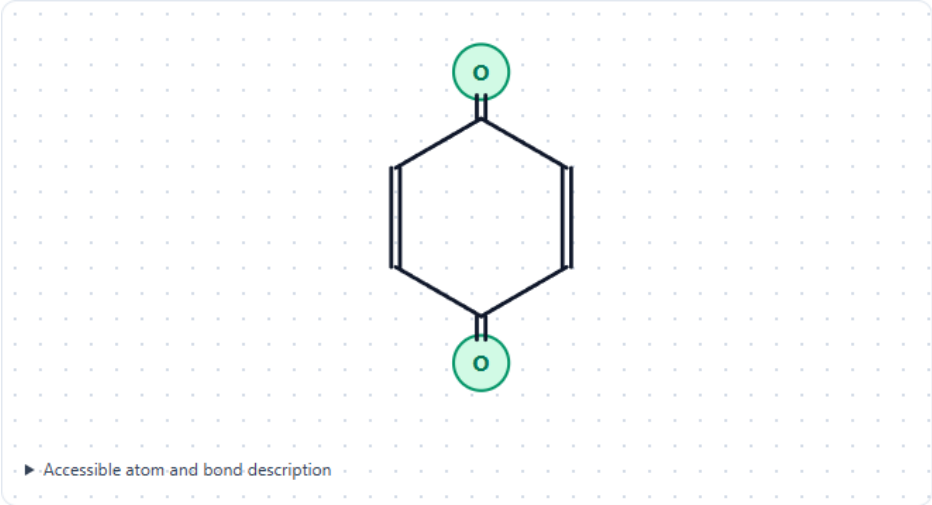
Answer type: choice

Option 0 (A): Reduction

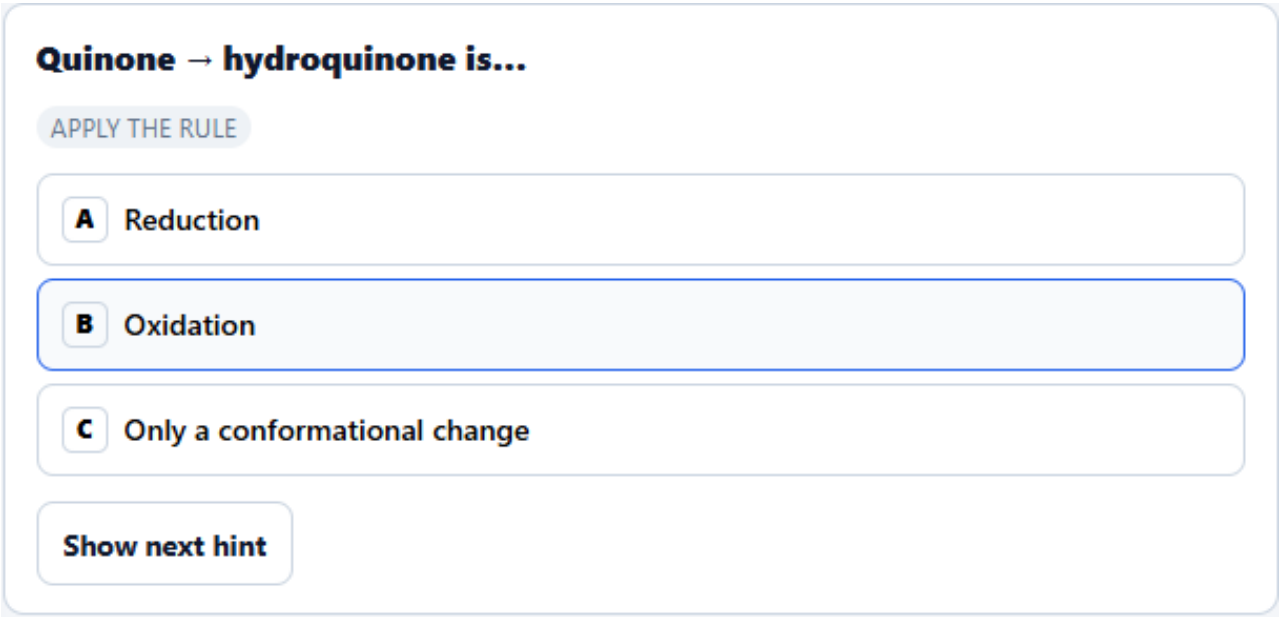
Option 1 (B): Oxidation

Option 2 (C): Only a conformational change

Screenshot: quinone-redox-stage-2-molecule



Screenshot: quinone-redox-stage-2-question



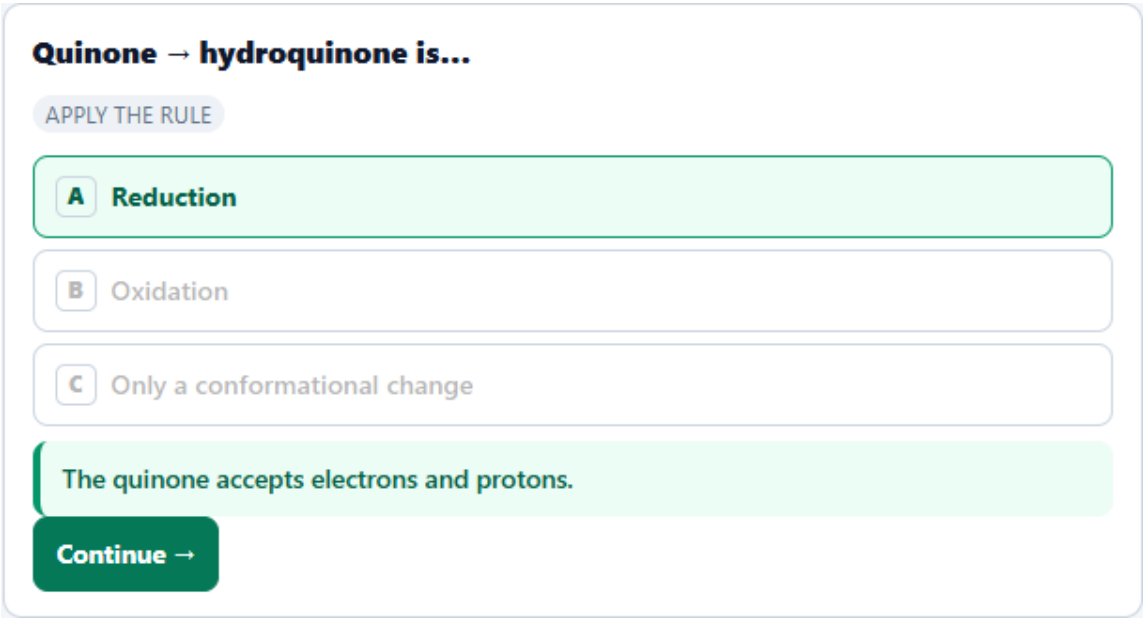
{"authorAnswer": 0, "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Track C=O versus C–OH.

Hint 2: The reduced form carries more hydrogen.

Explanation: The quinone accepts electrons and protons.

Screenshot: quinone-redox-stage-2-feedback

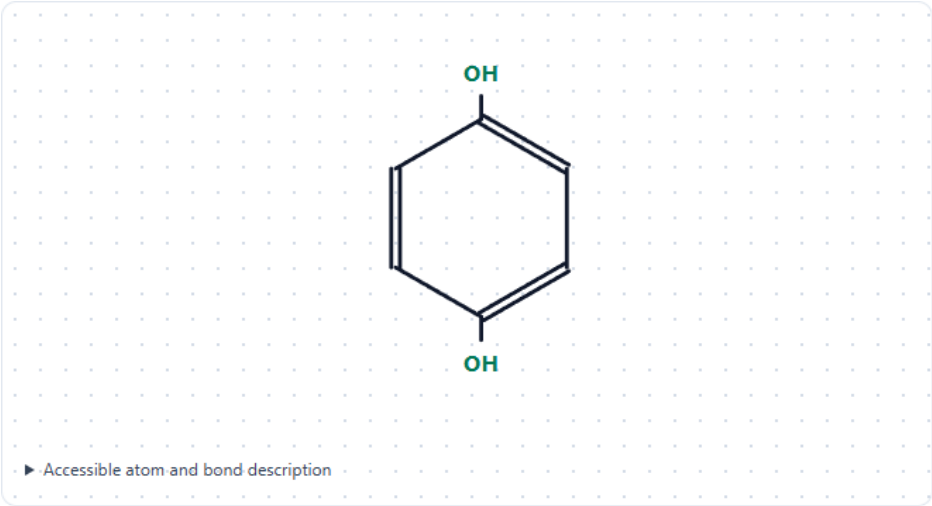


This stage uses a different molecule (connectivity shown).

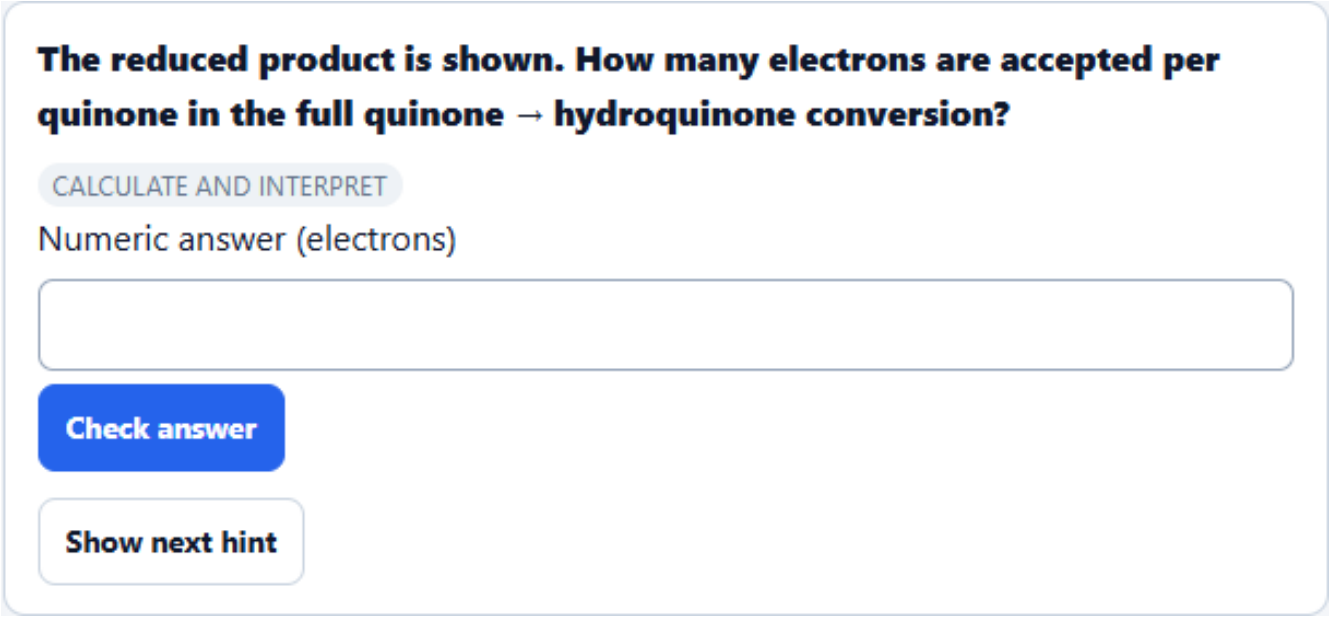
quinone-redox:stage-3 | The reduced product is shown. How many electrons are accepted per quinone in the full quinone → hydroquinone conversion?

Answer type: number | unit: electrons

Screenshot: quinone-redox-stage-3-molecule



Screenshot: quinone-redox-stage-3-question



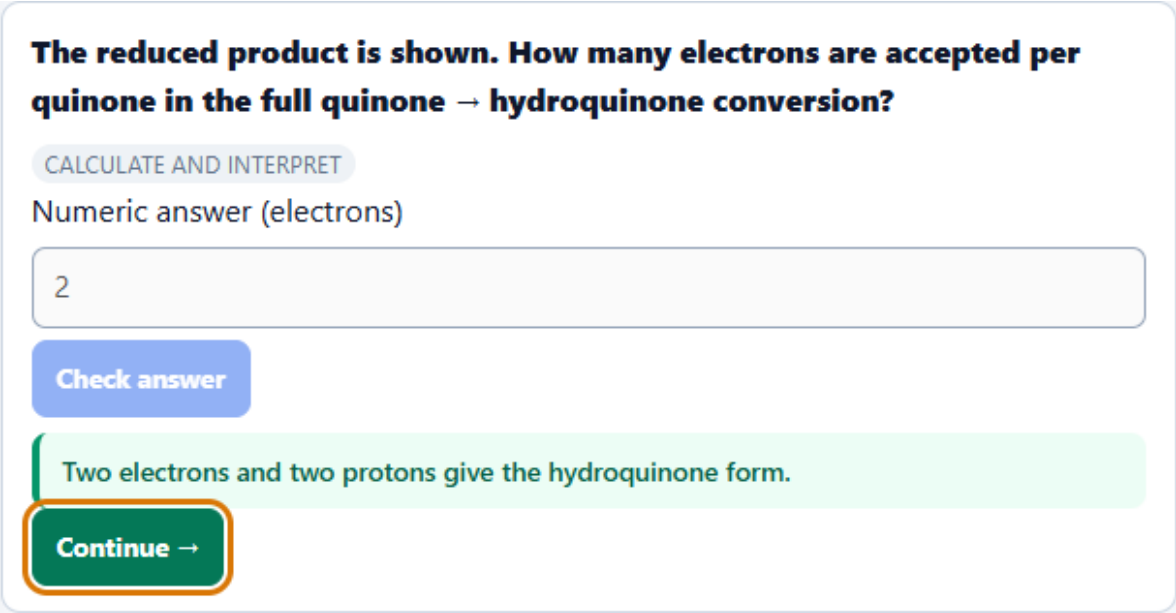
{"authorAnswer": 2, "tolerance": 0, "highlightsAfterCorrectAnswer": {}}

Hint 1: Compare the hydrogen counts of quinone and the reduced product.

Hint 2: Balance hydrogen using H+, then balance total charge using electrons.

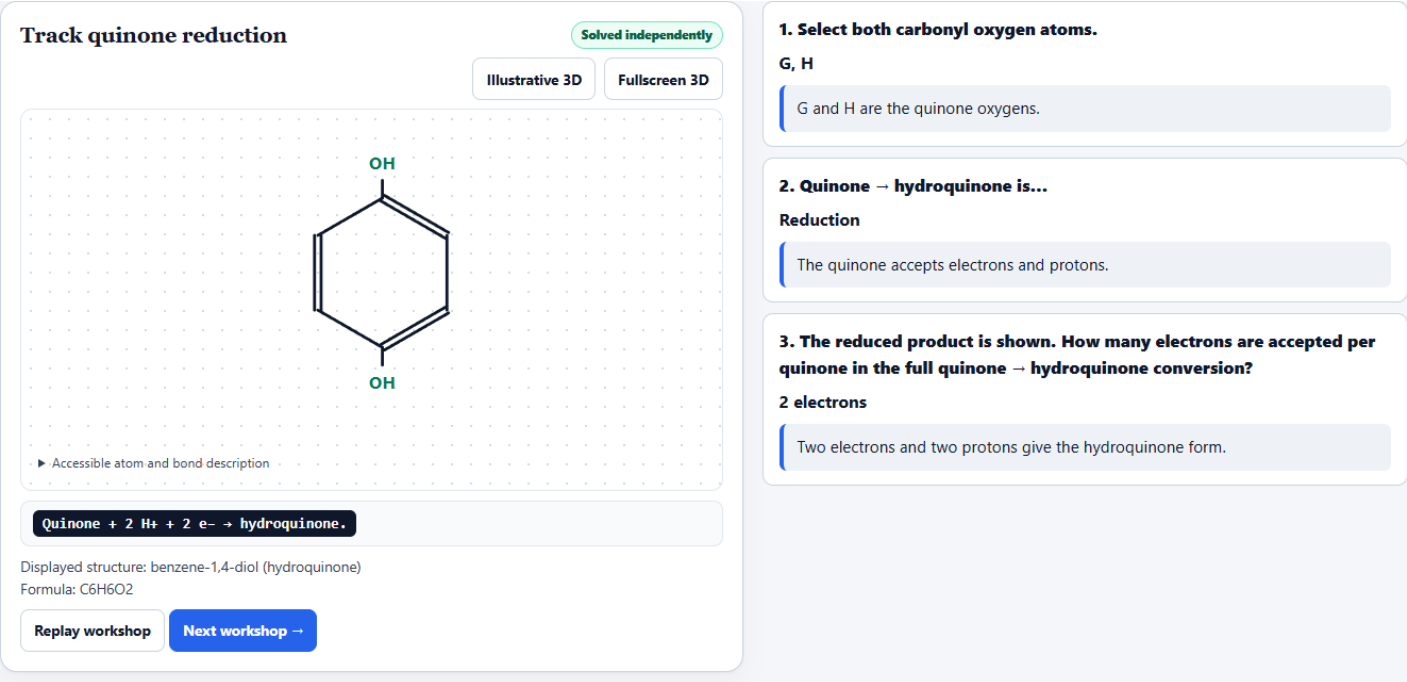
Explanation: Two electrons and two protons give the hydroquinone form.

Screenshot: quinone-redox-stage-3-feedback



Completion summary: Quinone + 2 H+ + 2 e− → hydroquinone.

Screenshot: quinone-redox-complete



Molecular repository: author claims to verify

isopropanol

Author claims: propan-2-ol | formula C3H8O | canonical composition C3H8O | net charge +0

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=50.00, y=115.00; charge=0; explicit H override=none

1 (B): C; x=108.00, y=85.00; charge=0; explicit H override=none

2 (C): C; x=166.00, y=115.00; charge=0; explicit H override=none

3 (D): O; x=108.00, y=25.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 1], [1, 3, 1]]

tbutanol

Author claims: 2-methylpropan-2-ol | formula C4H10O | canonical composition C4H10O | net charge +0

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=50.00, y=115.00; charge=0; explicit H override=none

1 (B): C; x=108.00, y=85.00; charge=0; explicit H override=none

2 (C): C; x=166.00, y=115.00; charge=0; explicit H override=none

3 (D): C; x=108.00, y=25.00; charge=0; explicit H override=none

4 (E): O; x=108.00, y=160.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 1], [1, 3, 1], [1, 4, 1]]

acetone

Author claims: propanone | formula C3H6O | canonical composition C3H6O | net charge +0

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=50.00, y=115.00; charge=0; explicit H override=none

1 (B): C; x=108.00, y=85.00; charge=0; explicit H override=none

2 (C): C; x=166.00, y=115.00; charge=0; explicit H override=none

3 (D): O; x=108.00, y=25.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 1], [1, 3, 2]]

acetaldehyde

Author claims: ethanal | formula C2H4O | canonical composition C2H4O | net charge +0

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=131.00, y=115.00; charge=0; explicit H override=none

1 (B): C; x=189.00, y=85.00; charge=0; explicit H override=none

2 (C): O; x=189.00, y=25.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 2]]

mesylate

Author claims: ethyl methanesulfonate | formula C3H8O3S | canonical composition C3H8O3S | net charge +0

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=35.00, y=130.00; charge=0; explicit H override=none

1 (B): C; x=85.00, y=100.00; charge=0; explicit H override=none

2 (C): O; x=140.00, y=130.00; charge=0; explicit H override=none

3 (D): S; x=200.00, y=100.00; charge=0; explicit H override=none

4 (E): C; x=255.00, y=130.00; charge=0; explicit H override=none

5 (F): O; x=170.00, y=40.00; charge=0; explicit H override=none

6 (G): O; x=230.00, y=40.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 1], [2, 3, 1], [3, 4, 1], [3, 5, 2], [3, 6, 2]]

acetal

Author claims: 2,2-dimethoxypropane | formula C5H12O2 | canonical composition C5H12O2 | net charge +0

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=75.00, y=130.00; charge=0; explicit H override=none

1 (B): C; x=135.00, y=100.00; charge=0; explicit H override=none

2 (C): C; x=195.00, y=130.00; charge=0; explicit H override=none

3 (D): O; x=105.00, y=45.00; charge=0; explicit H override=none

4 (E): O; x=165.00, y=45.00; charge=0; explicit H override=none

5 (F): C; x=50.00, y=15.00; charge=0; explicit H override=none

6 (G): C; x=220.00, y=15.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 1], [1, 3, 1], [1, 4, 1], [3, 5, 1], [4, 6, 1]]

phenol

Author claims: phenol | formula C6H6O | canonical composition C6H6O | net charge +0

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=170.00, y=37.00; charge=0; explicit H override=none

1 (B): C; x=228.89, y=71.00; charge=0; explicit H override=none

2 (C): C; x=228.89, y=139.00; charge=0; explicit H override=none

3 (D): C; x=170.00, y=173.00; charge=0; explicit H override=none

4 (E): C; x=111.11, y=139.00; charge=0; explicit H override=none

5 (F): C; x=111.11, y=71.00; charge=0; explicit H override=none

6 (G): O; x=170.00, y=5.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 2], [1, 2, 1], [2, 3, 2], [3, 4, 1], [4, 5, 2], [5, 0, 1], [0, 6, 1]]

hydroquinone

Author claims: benzene-1,4-diol (hydroquinone) | formula C6H6O2 | canonical composition C6H6O2 | net charge +0

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=170.00, y=37.00; charge=0; explicit H override=none

1 (B): C; x=228.89, y=71.00; charge=0; explicit H override=none

2 (C): C; x=228.89, y=139.00; charge=0; explicit H override=none

3 (D): C; x=170.00, y=173.00; charge=0; explicit H override=none

4 (E): C; x=111.11, y=139.00; charge=0; explicit H override=none

5 (F): C; x=111.11, y=71.00; charge=0; explicit H override=none

6 (G): O; x=170.00, y=5.00; charge=0; explicit H override=none

7 (H): O; x=170.00, y=205.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 2], [1, 2, 1], [2, 3, 2], [3, 4, 1], [4, 5, 2], [5, 0, 1], [0, 6, 1], [3, 7, 1]]

quinone

Author claims: 1,4-benzoquinone | formula C6H4O2 | canonical composition C6H4O2 | net charge +0

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=170.00, y=37.00; charge=0; explicit H override=none

1 (B): C; x=228.89, y=71.00; charge=0; explicit H override=none

2 (C): C; x=228.89, y=139.00; charge=0; explicit H override=none

3 (D): C; x=170.00, y=173.00; charge=0; explicit H override=none

4 (E): C; x=111.11, y=139.00; charge=0; explicit H override=none

5 (F): C; x=111.11, y=71.00; charge=0; explicit H override=none

6 (G): O; x=170.00, y=5.00; charge=0; explicit H override=none

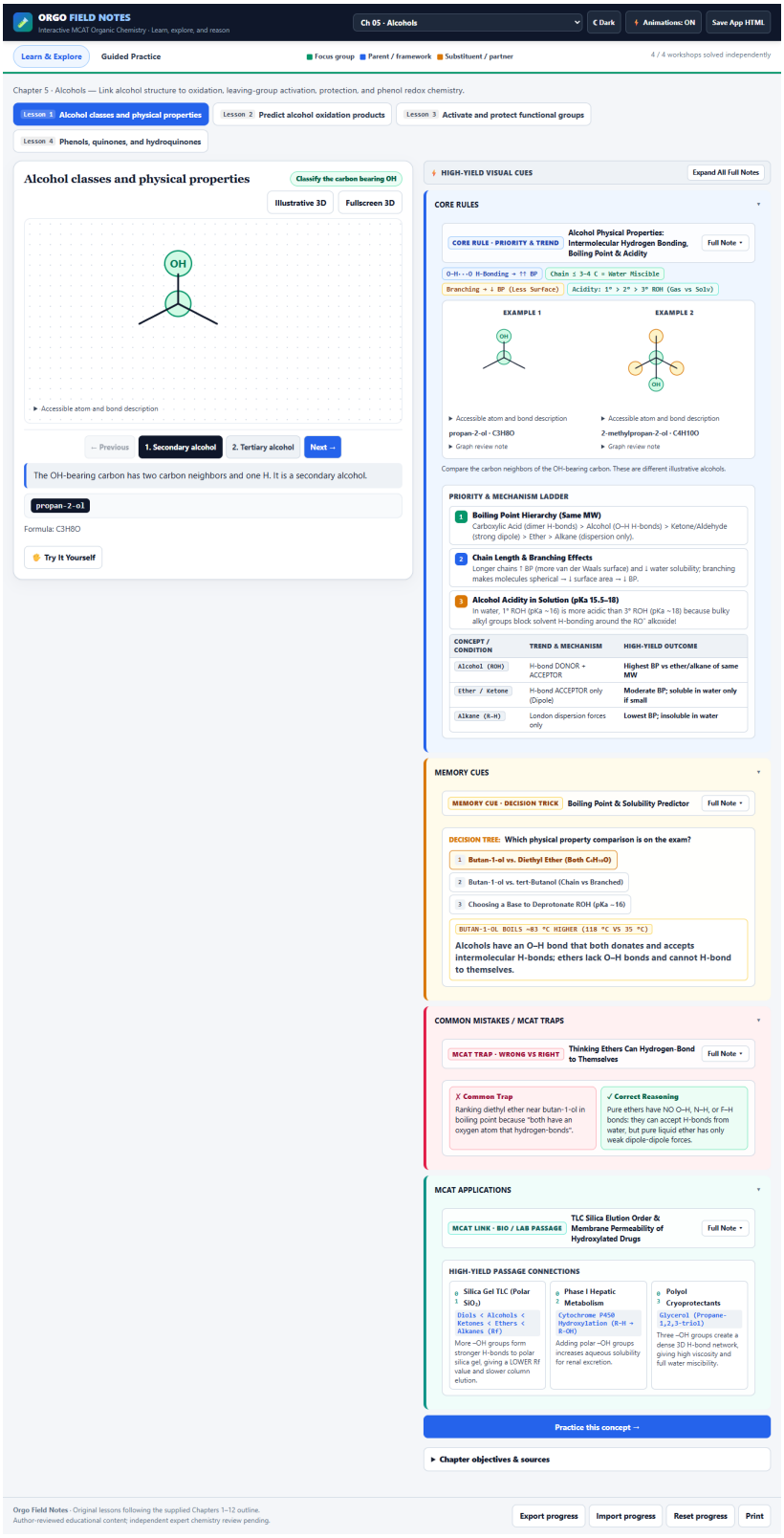
7 (H): O; x=170.00, y=205.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 2], [2, 3, 1], [3, 4, 1], [4, 5, 2], [5, 0, 1], [0, 6, 2], [3, 7, 2]]

Layout samples

These overview images are scaled for page fit. Detailed structure and stage images appear earlier. Samples cover the first concept on desktop and mobile in both themes, not every responsive state.

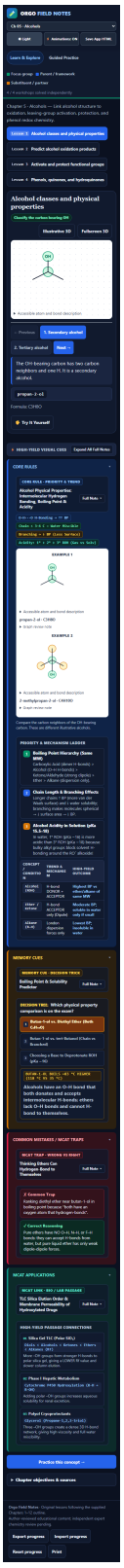
Screenshot: layout-desktop-light



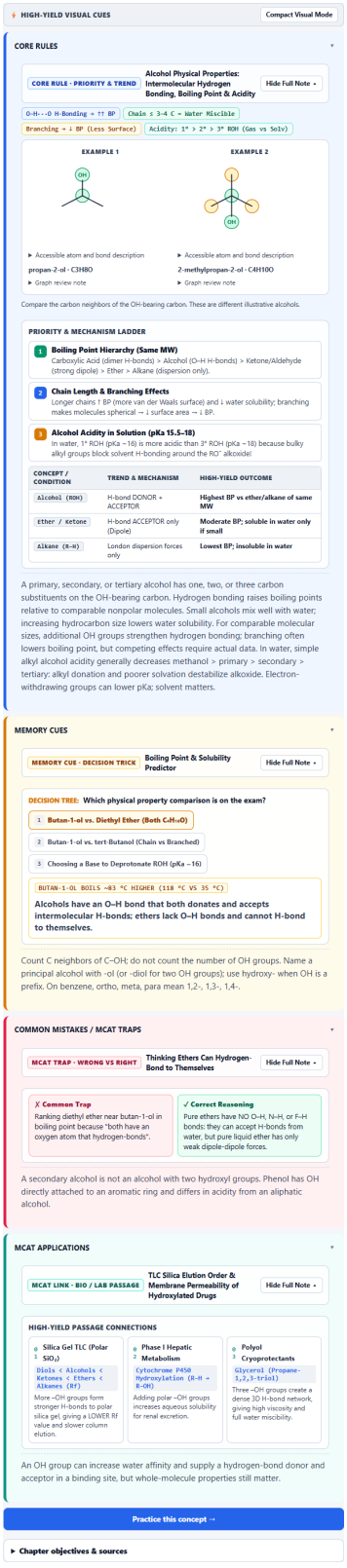
Screenshot: layout-mobile-light



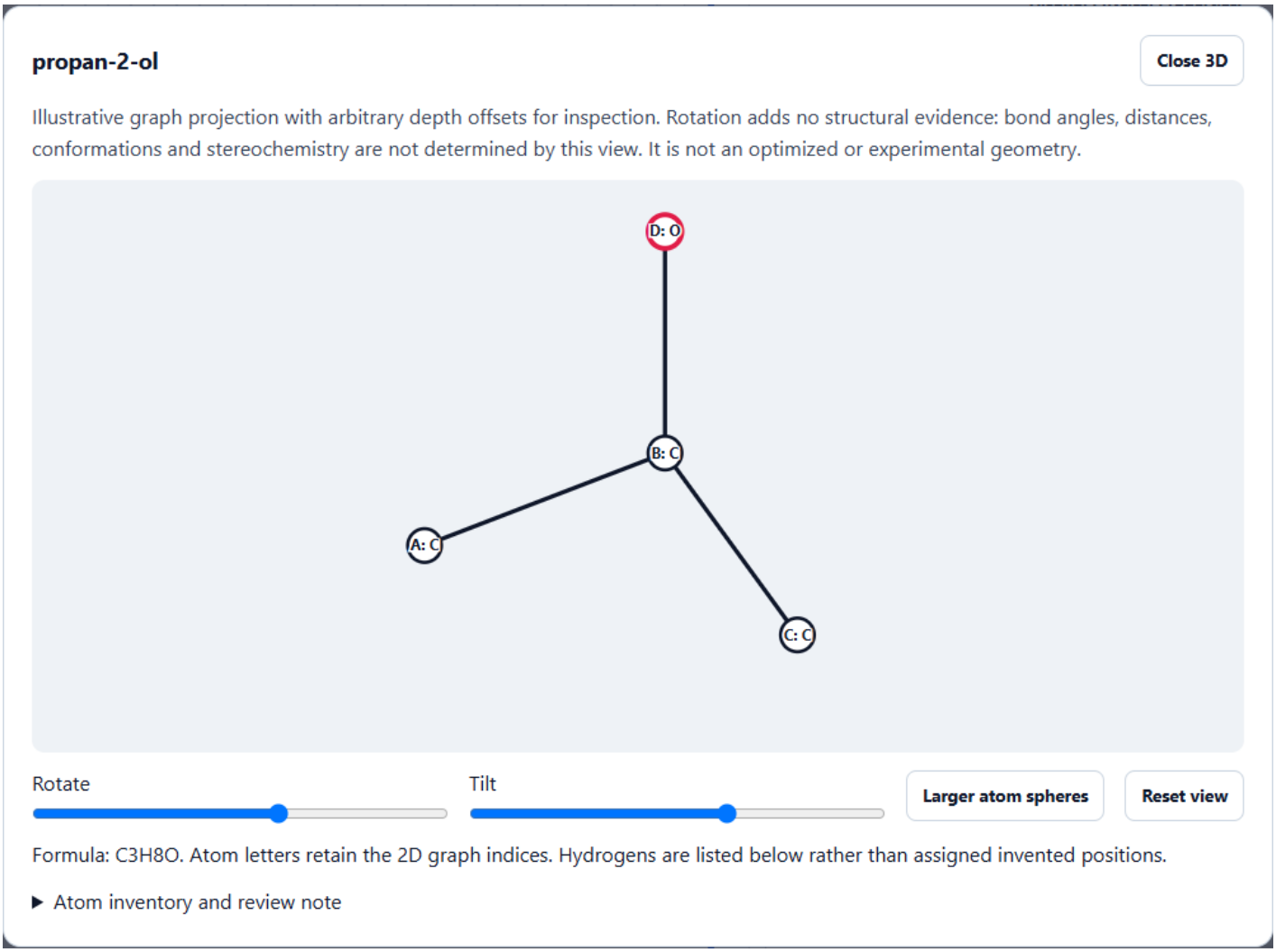
Screenshot: layout-mobile-dark



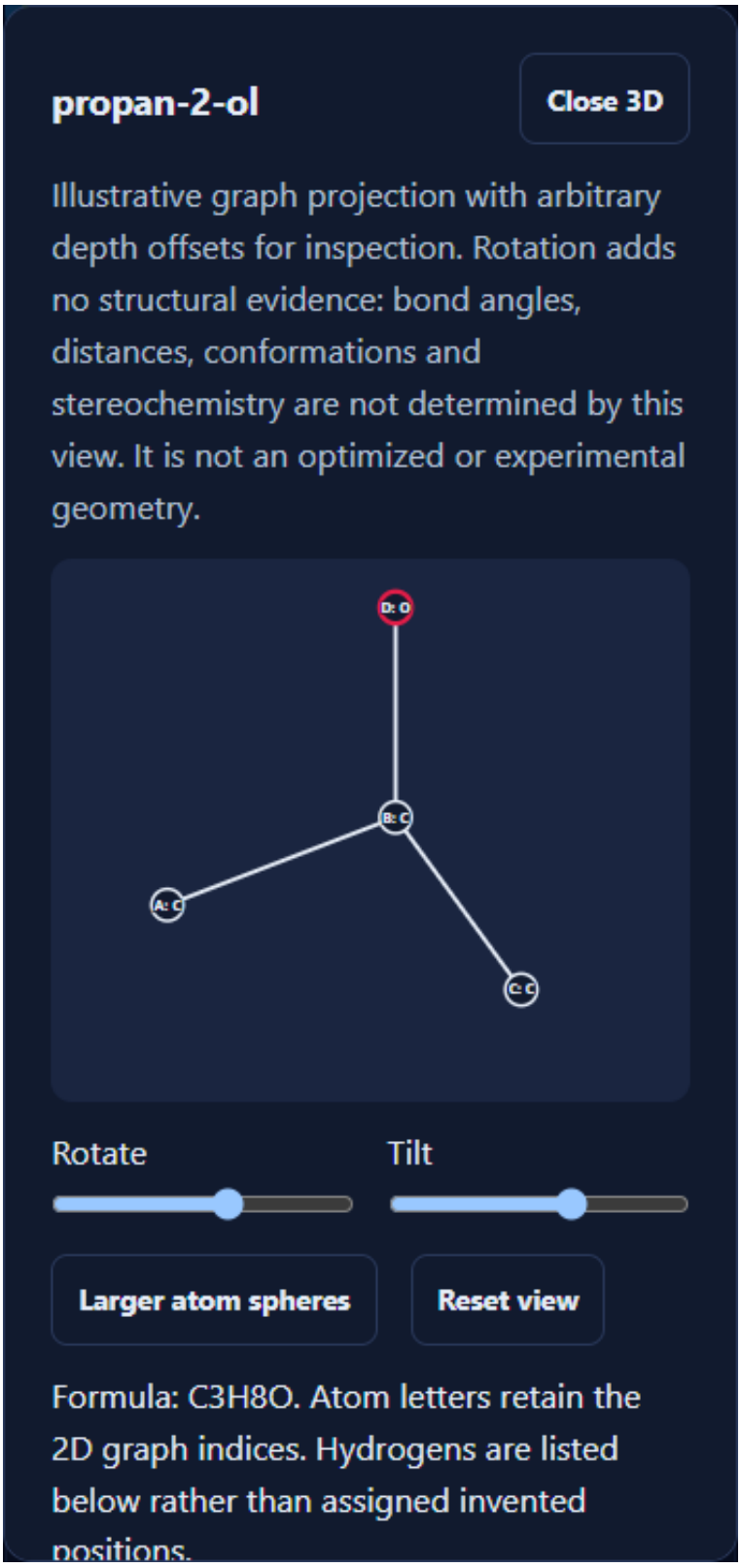
Screenshot: visual-notes-expanded



Screenshot: projection-desktop-light



Screenshot: projection-mobile-dark



Execution evidence

62 screenshots and 13 recorded interactions. Incorrect answers rejected and encoded keys accepted for captured guided stages. This checks implementation against its data; it does not validate chemistry independently. Static screenshots do not verify animation timing, arbitrary clicks, or screen-reader experience.